1 Problem and answer

Let $D \subseteq \mathbb{R}^d$, $d \geq 2$, be the Euclidean unit ball. For a Hölder–Zygmund conductivity $f \in C^\alpha(D)$, $\alpha > 1$, let $u = u_f$ solve the Darcy problem

$$\operatorname{div}(f \nabla u) = 1 \ \text{ on } D, \qquad u|_{\partial D} = 0.$$

We observe

$$Y \ = \ u_f \ + \ \frac{1}{\sqrt{n}}\, \mathbb{W},$$

where $\mathbb{W}$ is Gaussian white noise on $L^2(D)$, and we estimate f over the class

$$\mathcal{F}(\alpha, r) = \{\, h \in C^\alpha(D) : \ \|h - \mathbf{1}\|_{C^\alpha(D)} \leq r \,\}, \qquad r \in (0,1) \text{ small and fixed.}$$

(The class is contained in C^a with the a-th derivatives uniformly t-Hölder, where $\alpha = a + t$, $a = \lceil \alpha \rceil - 1 \in \mathbb{Z}_{\geq 1}$, $t \in (0,1]$, Zygmund if $t = 1$; in particular every $f \in \mathcal{F}(\alpha, r)$ is continuously differentiable and satisfies $1 - r \leq f \leq 1 + r$.)

Theorem 1.1. *There exist $r_0 = r_0(\alpha, d) \in (0,1]$ and, for every fixed $r \in (0, r_0]$, constants $c, C > 0$ with C depending only on (α, d) and c on (α, d, r), such that*

$$c\, \Phi_n \ \leq \ \inf_{\widehat{f}_n} \ \sup_{f_0 \in \mathcal{F}(\alpha, r)} \mathbb{E}_{f_0}\big[\|\widehat{f}_n - f_0\|^2_{L^2(D)}\big] \ \leq \ C\, \Phi_n, \qquad \Phi_n = n^{-\frac{2\alpha}{2\alpha + d + 2}}.$$

Answer. The minimax rate is

$$\boxed{n^{-\frac{2\alpha}{2\alpha+d+2}}} \qquad \text{up to logarithmic factors.}$$

Both bounds proved below carry this power of n; our argument is in fact log-free, which is a fortiori "up to logarithmic factors." This is *faster* than the direct white-noise rate $n^{-2\alpha/(2\alpha+d)}$: the denominator gains an additive $+2$, i.e. the inverse problem is only *mildly ill-posed of degree one*. The mechanism, proved below, is that the linearization T of $f \mapsto u_f$ at $\mathbf{1}$ satisfies $\Delta(Tg) = g + \frac{x \cdot \nabla}{d} g$, and the transport-type source $g \mapsto g + \frac{x \cdot \nabla}{d} g$ is invertible as an order-zero operator ($L^{-1} = J \in \mathcal{B}(L^2(D))$ with $\|J\| \leq 2d$, $L \in \mathcal{B}(H^1(D), L^2(D))$; Lemma 2.3); consequently T behaves like a general (nonlocal) operator of order -1, gaining exactly one derivative on scale-δ perturbations, rather than two.
The lower bound is Proposition 4.1; the upper bound is Proposition 5.1. Constants implicit in $\lesssim, \asymp$ depend only on (α, d) and a fixed bump, never on f, n, or δ; C is uniform in $r \in (0, r_0]$, while the lower-bound c carries the factor r^2 of the shrinking class (Proposition 4.1).

2 Preliminaries

2.1 White-noise model and likelihood

Gaussian white noise $\mathbb{W}$ on $L^2(D)$ is the centred Gaussian process indexed by $L^2(D)$ with $\mathbb{E}\langle \mathbb{W}, \phi \rangle = 0$ and $\mathbb{E}[\langle \mathbb{W}, \phi \rangle \langle \mathbb{W}, \psi \rangle] = \langle \phi, \psi \rangle_{L^2(D)}$. Under conductivity f, the law P_f of Y is the Gaussian shift with mean u_f and covariance $n^{-1} I$ on $L^2(D)$. For two conductivities f, f_0 the Cameron–Martin formula gives

$$\mathrm{KL}(P_f \,\|\, P_{f_0}) \ = \ \frac{n}{2}\, \|u_f - u_{f_0}\|^2_{L^2(D)}. \tag{1}$$

All norms $\|\cdot\|$ are $L^2(D)$ unless otherwise indicated; $\|\cdot\|_p$ denotes $L^p(D)$.

2.2 External tools (precise statements)

We collect the standard results used, with precise hypotheses, so that no nonstandard result enters tacitly.

(T1) **Fano's inequality.** For probability measures $P_1, \dots, P_N$ on a common space, any estimator $\widehat{\theta}$ of the index, and any reference $\bar{P}$, $\frac{1}{N}\sum_j P_j(\widehat{\theta} \neq j) \geq 1 - \frac{\frac{1}{N}\sum_j \mathrm{KL}(P_j \| \bar{P}) + \log 2}{\log N}$.

(T2) **Varshamov–Gilbert.** For $M \geq 1$ there exist $N \geq 2^{cM}$ subsets $S \subset \{1, \dots, M\}$, each of cardinality $M/2$, with pairwise Hamming distance $|S \triangle S'| \geq M/8$; $c > 0$ absolute.

(T3) **Weyl's law.** For the Dirichlet Laplacian on the bounded smooth domain D, $\#\{\lambda_\nu \leq \Lambda\} \sim c_d |D| \Lambda^{d/2}$; hence the span of the eigenfunctions with $\lambda_\nu \leq M^2$ has dimension $\asymp M^d$.

(T4) **Cameron–Martin.** For $Y = u + n^{-1/2}\mathbb{W}$ on $L^2(D)$, means $u_1, u_0 \in L^2(D)$ give laws with $\mathrm{KL}(P_{u_1} \| P_{u_0}) = \frac{n}{2}\|u_1 - u_0\|_2^2$.

(T5) **Wavelet/Besov characterization of C^α and square function on D.** There is a boundary-corrected wavelet frame of the smooth domain D (Cohen–Dahmen–DeVore-type) such that for $\alpha > 0$ (noninteger, with the Zygmund convention at integers), $f \in C^\alpha(D) = B^\alpha_{\infty,\infty}(D)$ iff $\sup_\lambda 2^{j_\lambda(\alpha + d/2)} |\langle f, \psi_\lambda \rangle| < \infty$, comparable to $\|f\|_{C^\alpha}$; and $\|f\|_{L^2} \asymp \big(\sum_j \|\Delta_j f\|_{L^2}^2\big)^{1/2}$ for the spectral dyadic pieces $\Delta_j = \varphi_j(\sqrt{-\Delta})$ (the Dirichlet-Laplacian smooth functional calculus / heat-kernel Mikhlin square-function inequality on the smooth bounded domain).

(T6) **Divergence-form elliptic regularity.** For $1 - r \leq f \leq 1 + r$, $f \in C^\alpha$, $\alpha > 1$, $\mathrm{div}(f \nabla u) = F$ in D with $u = 0$ on ∂D has a unique solution with $\|u\|_{C^{1,\beta}} \leq C(\alpha, d, r)\|F\|_{L^\infty}$ for some $\beta = \beta(\alpha, d) \in (0, 1]$; in particular $\|\nabla u\|_{L^\infty} \leq C$.

(T7) **Hardy / Cesàro.** The Cesàro operator $(Cf)(t) = \frac{1}{t}\int_0^t f(s)\, ds$ has norm $\frac{p}{p-1}$ ($= 2$ on L^2) on $L^p((0,1), dt)$.

Every PDE/functional bound below is either derived here or is an application of (T1)–(T7).

2.3 Linearization about the constant conductivity

The constant-conductivity solution is explicit:

$$u_0(x) = \frac{|x|^2 - 1}{2d}, \qquad \nabla u_0(x) = \frac{x}{d}, \qquad \Delta u_0 = 1.$$

For any $\phi \in C^1(D)$ one has the pointwise identity

$$\mathrm{div}(\phi \nabla u_0) = \nabla\phi \cdot \Big(\frac{x}{d}\Big) + \phi\, \Delta u_0 = \frac{x \cdot \nabla \phi}{d} + \phi. \tag{2}$$

Lemma 2.1 (Linearization)**.** *For $g \in L^2(D)$, let $Tg \in H^1_0(D)$ be the unique solution of $-\mathrm{div}(\nabla w) = -\mathrm{div}(g \nabla u_0)$, $w|_{\partial D} = 0$, i.e. $Tg = -\Delta^{-1}\big(\mathrm{div}(g \nabla u_0)\big)$ in the $H^{-1}(D)/H^1_0(D)$ sense. Then, as distributions,*

$$\Delta(Tg) = g + \frac{x \cdot \nabla}{d}\, g. \tag{3}$$

Proof. By (2), $\mathrm{div}(g \nabla u_0) = g + \frac{x \cdot \nabla}{d} g$ as a distribution of order ≤ 1, hence in $H^{-1}(D)$; applying Δ^{-1} gives (3). The Dirichlet Laplacian $-\Delta : H^1_0(D) \to H^{-1}(D)$ is an isomorphism (Lax–Milgram), and $\|Tg\|_{H^1} \lesssim \|g\|_{H^{-1}}$. □

Remark 2.2 (Transport structure of L). Write $A = \frac{x\cdot\nabla}{d}$ and $L = I + A$. Using $\operatorname{div}(xg) = (\operatorname{div} x)g + x \cdot \nabla g = dg + x \cdot \nabla g$ and $x \cdot \nu = |x| = 1$ on ∂D,

$$Lg = g + \frac{x \cdot \nabla g}{d} = \frac{1}{d}\big(dg + x \cdot \nabla g\big) = \frac{1}{d}\operatorname{div}(x\, g).$$

Hence L is a pure divergence composed with the vector field $x \mapsto x/d$; in particular *if* $\int_D g\,dx = 0$ *then* $\int_D Lg\,dx = \frac{1}{d}\int_{\partial D} g\,dS$ (this vanishes for the mean-zero wavelet packets used below since each ψ_m has integral zero and is interior). This divergence/transport structure makes L an order-zero invertible operator on $L^2(D)$ (Lemma 2.3); the smoothing order of $T = -\Delta^{-1}L$ is then governed by the dilation generator, giving exactly one degree (Lemmas 3.1, 3.3).

2.4 The transport operator L is an invertible order-zero operator

Lemma 2.3 (L is invertible, $\|L^{-1}\| \le C_d$)**.** *Let* $A = \frac{x\cdot\nabla}{d}$ *and* $L = I + A$. *For* $g \in C^1(D)$ *define*

$$(Jg)(x) \;:=\; d\int_0^1 \tau^{d-1} g(\tau x)\,d\tau, \qquad x \in D \setminus \{0\}.$$

Then:

(i) $LJg = JLg = g$ *on* $C^1(D)$, *so* L *is invertible and* $L^{-1} = J$.

(ii) $\|L^{-1}g\|_{L^2(D)} \le C_d\|g\|_{L^2(D)}$ *for every* $d \ge 2$ *(in fact* $C_d = 2d$ *suffices).*

(iii) Consequently $\|Lg\|_2 \ge c_d\|g\|_2$ *(*$c_d = C_d^{-1} > 0$*). Precisely:* $J = L^{-1}$ *extends to an element of* $\mathcal{B}(L^2(D))$, $L \in \mathcal{B}(H^1(D), L^2(D))$ *(since* $\|Lg\|_2 \le \|g\|_2 + \frac{1}{d}\|\nabla g\|_2$*). Moreover, writing* $Lg = \frac{1}{d}\operatorname{div}(xg)$ *and defining* $\operatorname{div}(xg) \in H^{-1}(D)$ *for general* $g \in L^2(D)$ *by duality, the operator* $T(g) := -\Delta^{-1}\big(\frac{1}{d}\operatorname{div}(xg)\big)$ *extends the* $\Delta w = Lg$ *map of Lemma 2.1 from* C^1 *to all of* $L^2(D)$, *with*

$$T \in \mathcal{B}\big(L^2(D), H_0^1(D)\big) \cap \mathcal{B}\big(H^{-1}(D), L^2(D)\big), \qquad \|T\|_{L^2\to L^2} \le C_d \cdot 2C_P,$$

in particular $\|Tg\|_2 \le C_T\|g\|_2$ *with* $C_T = C_T(d)$, *used in the spectral-tail step of Lemma 5.2.*

Proof. (i) Fix $x \neq 0$ and set $\phi(\tau) = g(\tau x)$, so $\phi'(\tau) = x \cdot \nabla g(\tau x)$. Then

$$\frac{d}{d\tau}\big[\tau^d g(\tau x)\big] = d\tau^{d-1} g(\tau x) + \tau^d x \cdot \nabla g(\tau x).$$

Integrating over $\tau \in [0,1]$ ($\tau^d g(\tau x) \to 0$ as $\tau \to 0$):

$$g(x) = \int_0^1 \Big(d\tau^{d-1} g(\tau x) + \tau^d x \cdot \nabla g(\tau x)\Big)d\tau$$

and, differentiating under the integral in x,

$$x \cdot \nabla(Jg)(x) = d\int_0^1 \tau^{d-1}\big(\tau x \cdot \nabla g(\tau x)\big)d\tau = d\int_0^1 \tau^d\, x \cdot \nabla g(\tau x)\,d\tau.$$

Adding, $d\,(Jg)(x) + x \cdot \nabla(Jg)(x) = d\int_0^1 \big(d\tau^{d-1} g(\tau x) + \tau^d x \cdot \nabla g(\tau x)\big)d\tau = d\,g(x)$ by the previous display; dividing by d, $L(Jg)(x) = (Jg)(x) + \frac{x\cdot\nabla(Jg)(x)}{d} = g(x)$, i.e. $LJ = I$. Applying L first and using the same integration with $(Lg)(\tau x) = g(\tau x) + \frac{\tau x\cdot\nabla g(\tau x)}{d}$ gives $JLg = g$; hence $L^{-1} = J$.

(ii) Polar coordinates $x = \rho\theta$, $dx = \rho^{d-1}d\rho\,d\sigma(\theta)$:

$$\|L^{-1}g\|_2^2 = \int_{\mathbb{S}^{d-1}}\int_0^1 \Big|d\int_0^1 \tau^{d-1} g(\tau\rho\theta)d\tau\Big|^2 \rho^{d-1}d\rho\,d\sigma(\theta).$$

Substitute $s=\tau\rho$: $d\int_0^1 \tau^{d-1}g(\tau\rho\theta)d\tau=\frac{d}{\rho^d}\int_0^\rho s^{d-1}g(s\theta)\,ds$. For fixed $\theta\in\mathbb{S}^{d-1}$ set $\gamma(\rho)=g(\rho\theta)$ and equip $(0,1)$ with the measure $d\mu(\rho)=\rho^{d-1}d\rho$, so that $\mu(0,\rho)=\rho^d/d$ and $\|g\|^2_{L^2(D)}=\int_{\mathbb{S}^{d-1}}\|\gamma\|^2_{L^2(d\mu)}d\sigma(\theta)$. The pointwise formula reads

$$(L^{-1}g)(\rho\theta)=\frac{d}{\rho^d}\int_0^\rho \gamma(s)\,d\mu(s)=d\,(H\gamma)(\rho),\qquad (H\gamma)(\rho):=\frac{1}{\mu(0,\rho)}\int_0^\rho \gamma\,d\mu.$$

The averaging operator H is bounded on $L^2(d\mu)$ (Hardy's inequality on $((0,1),d\mu)$). To compute a constant, change variables $t=\rho^d$: then $\rho^{d-1}d\rho=dt/d$, $\mu(0,\rho)=t/d$, and with $\beta(t):=d^{-1/2}\gamma(t^{1/d})$ one has the isometry $\int_0^1|\beta(t)|^2dt=\int_0^1|\gamma(\rho)|^2\rho^{d-1}d\rho=\|\gamma\|^2_{L^2(d\mu)}$, and

$$(H\gamma)(\rho)=\frac{d}{t}\int_0^{t^{1/d}}\gamma(s)\,s^{d-1}ds=\frac{d}{t}\int_0^t\gamma(s_1^{1/d})\,\frac{ds_1}{d}=\frac{\sqrt{d}}{t}\int_0^t\beta(s_1)\,ds_1,$$

which is $\sqrt{d}$ times the Cesàro operator in t (of norm $\frac{p}{p-1}=2$ on $L^2((0,1),dt)$). Therefore

$$\|H\gamma\|^2_{L^2(d\mu)}=\int_0^1|H\gamma|^2\frac{dt}{d}=\frac{1}{d}\cdot d\int_0^1\Big|\frac{1}{t}\int_0^t\beta\Big|^2dt\le\frac{d}{d}\cdot 4\int_0^1|\beta|^2dt=4\,\|\gamma\|^2_{L^2(d\mu)},$$

so $\|H\gamma\|_{L^2(d\mu)}\le 2\|\gamma\|_{L^2(d\mu)}$. Therefore

$$\|L^{-1}g\|^2_{L^2(D)}=d^2\int_{\mathbb{S}^{d-1}}\|H\gamma\|^2_{L^2(d\mu)}d\sigma\le d^2\cdot 4\int_{\mathbb{S}^{d-1}}\|\gamma\|^2_{L^2(d\mu)}d\sigma=4d^2\,\|g\|^2_{L^2(D)},\qquad \|L^{-1}g\|_2\le 2d\,\|g\|_2.$$

Thus $\|L^{-1}\|\le C_d$ with $C_d=2d$; any d-dependent constant suffices here, as all constants below are allowed to depend on (α,d). We thus take (ii) in the form $\|L^{-1}g\|_2\le C_d\|g\|_2$.

(iii) From $\|g\|_2=\|L^{-1}Lg\|_2\le C_d\|Lg\|_2$, $\|Lg\|_2\ge c_d\|g\|_2$. On a dense set (C^1) $J=L^{-1}$ with $\|J\|\le C_d$, so J extends to $\mathcal{B}(L^2(D))$; and $\|Lg\|_2\le\|g\|_2+\frac{1}{d}\|x\cdot\nabla g\|_2\le(1+\frac{1}{d})\|g\|_{H^1}$, so $L\in\mathcal{B}(H^1,L^2)$ (it is *not* bounded on L^2). For the last claim, map $g\in L^2$ to the divergence-form distribution $F=\frac{1}{d}\operatorname{div}(xg)\in H^{-1}$ ($\|F\|_{H^{-1}}\le\frac{1}{d}\|g\|_2$) and apply the Dirichlet Laplacian inverse: $-\Delta^{-1}:H^{-1}\to H^1_0$ is bounded, so $T:L^2\to H^1_0$ is bounded with $\|Tg\|_{H^1}\lesssim\frac{1}{d}\|g\|_2$; by Poincaré $\|Tg\|_2\lesssim\|g\|_2$. Finally, for $\psi\in L^2$ and $\phi=-\Delta^{-1}\psi\in H^2\cap H^1_0$ ($\|\phi\|_{H^2}\lesssim\|\psi\|_2$), $|\langle Tg,\psi\rangle|=\frac{1}{d}|\int_D g\,x\cdot\nabla\phi\,dx|\le\frac{1}{d}\|g\|_{H^{-1}}\|x\cdot\nabla\phi\|_{H^1_0}\lesssim\|g\|_{H^{-1}}\|\psi\|_2$, so $T:H^{-1}\to L^2$ is bounded. On $g\in C^1$ the identity $\frac{1}{d}\operatorname{div}(xg)=Lg$ identifies this extension with the T of Lemma 2.1; in particular Tg always satisfies the very-weak form of $\Delta w=Lg$. The refined scale-localized bound $\|Tg\|_2\lesssim\delta\|g\|_2$ for packets is Lemma 3.3. □

Remark 2.4. Lemma 2.3 is the analytic crux: by Lemma 2.1, $Tg=-\Delta^{-1}Lg$. On a frequency band $|\cdot|\asymp\delta^{-1}$ the Dirichlet Laplacian inverts with gain δ^2 but the *transport source* L is invertible as an order-zero operator (Lemma 2.3: $L^{-1}=J\in\mathcal{B}(L^2)$ and $\|Lg\|_2\ge c_d\|g\|_2$), so T gains only *one* power of δ. This is made precise in Lemma 3.3.

Lemma 2.5 (Elliptic estimate for the harmonic part)**.** *Let $F\in H^{-1}(D)$ and decompose the solution $w\in H^1_0(D)$ of $-\Delta w=F$, $w|_{\partial D}=0$, as $w=U-q$ with $U=(-\Delta_{\mathbb{R}^d})^{-1}F\in H^1_{\mathrm{loc}}$ (fundamental-solution representation on the compactly supported F) and $\Delta q=0$ in D, $q=U$ on ∂D. Then*

$$\|q\|_{L^2(D)}\ \le\ C_E\,\|U\|_{H^1(D)},\qquad C_E=C_E(d),$$

and consequently, for $F=\frac{1}{d}\operatorname{div}(xg)$ ($g\in L^2$) and $w=Tg$, $\|q\|_2\le C''_E\|g\|_2$ with $C''_E=C''_E(d)$. If moreover $\operatorname{supp}g\subset D_{(\tau)}:=\{|x|\le 1-\tau\}$ *for some $\tau\in(0,\frac{1}{2}]$, then with $C'''_E=C'''_E(d)$,*

$$\|U\|_{H^1(\partial D)}\ \le\ C'''_E\,\tau^{-(d-1)}\|g\|_2,$$

*i.e. the harmonic trace stays $O(\|g\|_2)$ (up to a τ-*fixed *factor) and is not amplified as the source recedes from ∂D.*

Proof. Set $\phi := q - U$, the solution of $-\Delta\phi = F$ in $H^1_0(D)$ (since $\Delta q = 0$, $-\Delta U = F$, and $\phi = q - U = 0$ on ∂D); H^1_0 energy theory (Lax–Milgram and Poincaré) gives $\|\phi\|_{H^1} \le C_P \|F\|_{H^{-1}}$. Testing against ϕ,

$$\int_D |\nabla\phi|^2 = \langle F, \phi\rangle_{H^{-1}, H^1_0} \le \|F\|_{H^{-1}} \|\phi\|_{H^1},$$

so $\|\phi\|_{H^1} \le C_P\|F\|_{H^{-1}}$, and $\|q\|_{L^2} \le \|U\|_{L^2} + \|\phi\|_{L^2} \le \|U\|_{L^2} + C_P\|F\|_{H^{-1}} \le C_E\|U\|_{H^1}$ after using $\|F\|_{H^{-1}} \lesssim \|U\|_{H^1}$. For the specific $F = \frac{1}{d}\operatorname{div}(xg)$, $\|U\|_{H^1(D)} \lesssim \|F\|_{H^{-1}} \le \frac{1}{d}\|g\|_2$ (interior elliptic regularity on $D \Subset \frac{3}{2}D$ for $-\Delta_{\mathbb{R}^d} U = F$, plus $\|F\|_{H^{-1}} \le \frac{1}{d}\|g\|_2$ from the divergence form), so $\|q\|_2 \le C''_E\|g\|_2$. For the interior claim, use the representation $U = (-\Delta_{\mathbb{R}^d})^{-1}\big(\frac{1}{d}\operatorname{div}(xg)\big) = -\frac{1}{d}(-\Delta_{\mathbb{R}^d})^{-1}\operatorname{div}(xg)$, i.e. $U(x) = -\frac{1}{d}\int \nabla\Phi(x-y)\cdot y\, g(y)\, dy$ with Φ the fundamental solution, $|\nabla\Phi(z)| \lesssim |z|^{1-d}$ (integration by parts is unnecessary: this is the convolution of the L^2 field yg with the locally-integrable kernel $\nabla\Phi$). For $x \in D\setminus D_{(\tau/2)}$ and $y \in D_{(\tau)}$, $|x-y| \ge \tau/2$, so $|\nabla\Phi(x-y)| \lesssim \tau^{1-d}$ and $|U(x)| \lesssim \tau^{1-d}\|g\|_1 \lesssim \tau^{1-d}\|g\|_2$, and the same with one more derivative, $|\nabla U(x)| \lesssim \tau^{-d}\|g\|_2$. Hence for a fixed interior/annular collar $D \setminus D_{(\tau/2)}$, $\|U\|_{H^1(D\setminus D_{(\tau/2)})} \lesssim \tau^{1-d}\|g\|_2$ and by smooth continuation (interior elliptic regularity on $D_{(\tau/2)}$ for the same equation, applied to the patched field) $\|U\|_{H^1(D)} \lesssim \tau^{1-d}\|g\|_2$; the trace theorem $\|u\|_{L^2(\partial D)} \lesssim \|u\|_{H^1(D)}$ then gives $\|U\|_{L^2(\partial D)} \lesssim \tau^{1-d}\|g\|_2$ with ∇U similarly bounded, so $\|U\|_{H^1(\partial D)} \lesssim \tau^{1-d}\|g\|_2$. The stated exponent $\tau^{-(d-1)}$ is not optimal but suffices: any (τ, d)-dependent constant works, since τ is chosen before δ_0. □

2.5 The forward map: very-weak form, apriori bounds, quadratic remainder

Since $f \in C^\alpha$, $\alpha > 1$, f is C^1 and $1 - r \le f \le 1 + r$ uniformly. We use the very-weak (L^2) formulation: $u_f \in L^2(D)$ satisfies, for all $\phi \in H^2(D) \cap H^1_0(D)$,

$$\int_D u_f \operatorname{div}(f\nabla\phi)\, dx = \int_D \phi\, dx. \tag{4}$$

Existence/uniqueness of a very-weak solution follows by transposition: since $f \in C^1$ and $1 - r \le f \le 1 + r$, the operator $-\operatorname{div}(f\nabla\cdot) : H^1_0(D) \to H^{-1}(D)$ is an isomorphism (Lax–Milgram), so for $\psi \in L^2(D)$ the adjoint problem $-\operatorname{div}(f\nabla\phi) = \psi$, $\phi \in H^1_0(D)$, has a unique solution $\phi \in H^2(D) \cap H^1_0(D)$; defining $u_f \in L^2(D)$ by $\langle u_f, \psi\rangle := \int_D \phi\, dx$ (well-defined, linear, bounded in ψ since $\|\phi\|_{H^2} \lesssim \|\psi\|_2$) gives (4) upon $\operatorname{div}(f\nabla\phi) = -\psi$, and uniqueness in L^2 follows from the same isomorphy (if $u \in L^2$ satisfies (4) with the right side 0, the density of the adjoint solutions ψ in L^2 forces $u = 0$). All subsequent L^2/weak estimates use only ellipticity and boundedness via (T6); we never invoke $u_f \in H^1_0(D)$ from (4).

Lemma 2.6 (Apriori estimates)**.** *Uniform in $f \in \mathcal{F}(\alpha, r)$: (i) $\|u_f\|_\infty + \|u_f\|_2 \lesssim 1$; (ii) $\|u_f - u_0\|_2 \lesssim \|f - \mathbf{1}\|_2 \le r$; (iii) more generally $\|u_f - u_0\|_2 \le C_A\|(f-\mathbf{1})\nabla u_0\|_{H^{-1}(D)}$ with $C_A = C_A(d, r)$, and in particular $\|u_f - u_0\|_2 \lesssim \|f - \mathbf{1}\|_\infty$; if moreover $\operatorname{supp}(f - \mathbf{1}) \subset D'$ with $\operatorname{dist}(D', \partial D) > \varepsilon_0$, then $\|u_f - u_0\|_2 \lesssim_{\varepsilon_0} \|f - \mathbf{1}\|_{H^1(D')}$.*

Proof. (i) By the comparison principle with barriers $\frac{u_0}{1\mp r} = \frac{|x|^2-1}{2d(1\mp r)}$: since $1 - r \le f \le 1 + r$ and $\operatorname{div}(f\nabla u_f) = 1$, the weak maximum principle for the divergence operator $w \mapsto \operatorname{div}(f\nabla w)$ (with the subsolution/supersolution comparison valid in the weak sense) gives, comparing u_f against the two radial barriers, $\operatorname{div}\big(f\nabla\frac{u_0}{1+r}\big) = \frac{f}{1+r}\Delta u_0 = \frac{f}{1+r} \le 1 = \operatorname{div}(f\nabla u_f)$ with $\frac{u_0}{1+r} = 0 = u_f$ on ∂D, so $u_f \ge \frac{u_0}{1+r}$ in D; symmetrically $\operatorname{div}\big(f\nabla\frac{u_0}{1-r}\big) = \frac{f}{1-r} \ge 1$ gives $u_f \le \frac{u_0}{1-r}$. Hence $\frac{|x|^2-1}{2d(1+r)} \le u_f \le \frac{|x|^2-1}{2d(1-r)} \le 0$ and $|u_f| \le \frac{1}{2d(1-r)} \lesssim 1$ on D. (ii) Testing (4) for u_f and u_0 against $\phi = -\Delta^{-1}\psi$ ($\psi \in L^2$ arbitrary, $\|\phi\|_{H^2} \lesssim \|\psi\|_2$) and subtracting:

$$\langle u_f - u_0, \psi\rangle = -\int_D (f - \mathbf{1})\nabla u_0 \cdot \nabla\phi\, dx,$$

so $|\langle u_f - u_0, \psi\rangle| \le \|f-\mathbf{1}\|_2 \|\nabla u_0\|_\infty \|\nabla\phi\|_2 \lesssim \|f-\mathbf{1}\|_2\|\psi\|_2$, and (ii) follows by duality. (iii) Sharpening the same display, $|\langle u_f - u_0, \psi\rangle| \le \|(f-\mathbf{1})\nabla u_0\|_{H^{-1}} \|\nabla\phi\|_{H^1_0} \le C\|(f-\mathbf{1})\nabla u_0\|_{H^{-1}}\|\psi\|_2$, giving $\|u_f - u_0\|_2 \lesssim \|(f-\mathbf{1})\nabla u_0\|_{H^{-1}}$. Now $(f-\mathbf{1})\nabla u_0 = (f-\mathbf{1})x/d$. On the full ball $\|(f-\mathbf{1})x/d\|_{H^{-1}} \le \|(f-\mathbf{1})x/d\|_2 \lesssim \|f-\mathbf{1}\|_\infty$ (since H^{-1} is weaker than L^2 via the embedding of the source as a divergence). If $\operatorname{supp}(f-\mathbf{1}) \subset D' \Subset D$, fix a cut-off $\chi_0 \in C_c^\infty(D)$, $\chi_0 = 1$ on D', and compute $\operatorname{div}\big(\chi_0(f-\mathbf{1})x/d\big) = (f-\mathbf{1})\frac{x\cdot\nabla\chi_0}{d} + \chi_0\big(f - \mathbf{1} + \frac{x\cdot\nabla f}{d}\big)$; this is an H^{-1} function of norm $\lesssim \|f-\mathbf{1}\|_{L^2(D')} + \|\nabla f\|_{L^2(D')}$, and $\|(f-\mathbf{1})x/d\|_{H^{-1}} = \|\chi_0(f-\mathbf{1})x/d\|_{H^{-1}} \lesssim \|f-\mathbf{1}\|_{H^1(D')}$ by Lax–Milgram duality (the map $g \mapsto \operatorname{div}(\chi_0 g x/d)$ is bounded $H^1_0(D) \to H^{-1}(D)$ and for g supported in D', $\chi_0 g = g$). (The gradient term ∇f is present because the conductivity does *not* vanish on ∂D; without the interior-support hypothesis the map $f \mapsto u_f$ is $C^\alpha \mapsto L^2$ bounded only with the additive $C(f-\mathbf{1})$ constant of (ii), which is consistent with – and does not improve – the radius minimax rate.) □

Lemma 2.7 (Quadratic remainder)**.** *For $f = \mathbf{1} + h \in \mathcal{F}(\alpha, r)$, write $u_f - u_0 = Th + R(h)$. Then $\|R(h)\|_2 \le C\|h\|_\infty\|h\|_2$.*

Proof. From the equations for u_f and u_0, $w := u_f - u_0$ satisfies $\operatorname{div}(f\nabla w) = -\operatorname{div}(h\nabla u_0)$, $w = 0$ on ∂D. The energy identity (test against w, justified since $f \in C^\alpha$, $\alpha > 1$ gives $w \in H^1$) gives $(1-r)\|\nabla w\|_2^2 \le \int_D f|\nabla w|^2 = -\int_D h\nabla u_0 \cdot \nabla w \le \|h\|_2\|\nabla u_0\|_\infty\|\nabla w\|_2$, hence $\|\nabla w\|_2 \lesssim \|h\|_2$. Since $R(h) = -\Delta^{-1}\operatorname{div}(h\nabla w)$, for any $\psi \in L^2$ with $\phi = -\Delta^{-1}\psi$, $\langle R(h), \psi\rangle = -\int_D h\nabla w\cdot\nabla\phi$, so $|\langle R(h),\psi\rangle| \le \|h\|_\infty\|\nabla w\|_2\|\nabla\phi\|_2 \lesssim \|h\|_\infty\|h\|_2\|\psi\|_2$. Duality gives the bound. □

Remark 2.8 ($\alpha > 1$ is essential). At the minimax scale $\delta_* \asymp n^{-1/(2\alpha+d+2)}$ the wavelet perturbations used below have $\|h\|_\infty \lesssim \delta_*^\alpha$, so $\|R(h)\|_2/(\delta_*\|h\|_2) \lesssim \delta_*^{\alpha-1} \to 0$ *iff* $\alpha > 1$. Thus the linearization is faithful precisely in the regime $\alpha > 1$, which is why the hypothesis appears.

3 Degree-one smoothing of the linearization

We now make Remark 2.4 precise with an exact Fourier-side computation, which shows T is an operator of order exactly -1.

Lemma 3.1 (Exact smoothing; Fourier form)**.** *Let $\chi \in C_c^\infty(\mathbb{R}^d)$, $0 \le \chi \le 1$, $\chi = 1$ on $\{|x| \le \frac12\}$, $\operatorname{supp}\chi \subset \{|x| \le \frac34\}$. For $h \in C^\alpha(D)$ set $\tilde h = h\chi$, extended by 0 to $\mathbb{R}^d$. Then, as tempered distributions,*

$$\widehat{L\tilde h}(\xi) = -\frac{1}{d}\,\xi\cdot\nabla_\xi\widehat{\tilde h}(\xi). \tag{5}$$

Consequently, for a scale-δ wavelet packet $\tilde h$ *and dyadic shells $S_k = \{\xi : |\xi| \in [2^k, 2^{k+1})\}$,*

$$\| |\xi|^{-2}\widehat{L\tilde h}(\xi)\|_{L^2(S_k)} = \frac{1}{d}\,\||\xi|^{-2}\,\xi\cdot\nabla_\xi\widehat{\tilde h}(\xi)\|_{L^2(S_k)}, \qquad \|U\tilde h\|_{L^2(S_k)} \asymp \delta\|\tilde h\|_{L^2(S_k)} \text{ on the loaded shell } 2^k \asymp \delta^{-1}, \tag{6}$$

the two-sided bound being valid for packets *(the situation in which the lower bound is used below; it is* not *a valid two-sided identity for an arbitrary C^α function, whose $\widehat{\tilde h}$ can be radially homogeneous and radially annihilate $\xi\cdot\nabla_\xi$).*

Proof. Using $\widehat{x_jF}(\xi) = i\,\partial_{\xi_j}\widehat F(\xi)$ and $\widehat{\partial_j h}(\xi) = i\xi_j\widehat h(\xi)$,

$$\widehat{x_j\partial_j\tilde h}(\xi) = i\,\partial_{\xi_j}\big(i\xi_j\widehat{\tilde h}(\xi)\big) = -\partial_{\xi_j}\big(\xi_j\widehat{\tilde h}(\xi)\big) = -\widehat{\tilde h}(\xi) - \xi_j\partial_{\xi_j}\widehat{\tilde h}(\xi).$$

Summing over $j = 1,\dots,d$: $\widehat{A\tilde h}(\xi) = \frac1d\sum_j \widehat{x_j\partial_j\tilde h}(\xi) = -\frac1d\big(d\,\widehat{\tilde h}(\xi) + \xi\cdot\nabla_\xi\widehat{\tilde h}(\xi)\big)$. Hence

$$\widehat{L\tilde h}(\xi) = \widehat{\tilde h}(\xi) + \widehat{A\tilde h}(\xi) = -\frac{\xi\cdot\nabla_\xi\widehat{\tilde h}(\xi)}{d}.$$

This proves (5). For the packet corollary (6) of a scale-δ packet $\tilde{h}(x) = \sum_m \varepsilon_m a_m \psi((x - x_m)/\delta)$, one expands $\widehat{\tilde{h}}(\xi) = \delta^d \widehat{\psi}(\delta\xi) \sum_m \varepsilon_m a_m e^{-ix_m\cdot\xi}$ and reads off the loaded-shell gain: $\xi \cdot \nabla_\xi$ differentiates the phase $e^{-ix_m\cdot\xi}$ to bring down a factor $x_m\cdot\xi \asymp |x_m|\delta^{-1}$, and on $|x_m| \geq c_0$ this is $\asymp \delta^{-1}$, so $\|\xi\cdot\nabla_\xi \widehat{\tilde{h}}\|_{L^2(S_k)} \asymp \delta^{-1}\|\widehat{\tilde{h}}\|_{L^2(S_k)}$; dividing by $|\xi|^2 \asymp \delta^{-2}$ gives (6) (with the phase interference controlled by separation and Lemma 3.2). The *lower* side of this packet estimate in the uniform sense relevant for (8) is reproved, with constants, inside Lemma 3.3's proof (the display labelled "Lower bound" there); the present form (6) records the identity content. □

Lemma 3.2 (Almost-orthogonality of separated exponentials)**.** *Let $\Lambda = \{x_m\} \subset \{|x| \leq \frac{1}{2}\}$ be 4δ-separated, $|x_m - x_{m'}| \geq 4\delta$. For $b = (b_m) \in \ell^2(\Lambda)$ and $\mathbf{a}_b(\xi) := \sum_m b_m e^{-ix_m\cdot\xi}$, define the frequency-ball average $\mathcal{M}_\rho(b) := \frac{1}{|B_\rho|}\int_{B_\rho} |\mathbf{a}_b(\xi)|^2 d\xi$ on $B_\rho = \{|\xi| \leq \rho\}$, $\rho = \Theta\delta^{-1}$, Θ a fixed constant. Then, uniformly in δ and the configuration Λ,*

$$\mathcal{M}_\rho(b) \lesssim \sum_m |b_m|^2, \qquad \text{and} \qquad \mathcal{M}_\rho(\{x_m\cdot(\cdot)\}\, b) \lesssim \delta^{-2}\sum_m |b_m|^2 \tag{7}$$

where the second statement applies to the weighted coefficients $\widetilde{b}_m = x_m \cdot v\, b_m$ with a fixed unit vector v (so $|\widetilde{b}_m| \leq |b_m||x_m| \leq \frac{1}{2}|b_m|$), and the corresponding phase function is $(x_m \cdot v)e^{-ix_m\cdot\xi}$.

Proof. Write, with $\chi_\rho = \mathbf{1}_{B_\rho}$ and $\check{\chi}_\rho$ its inverse Fourier transform, $\int_{B_\rho}|\mathbf{a}_b|^2 d\xi = \sum_{m,m'} b_m\overline{b_{m'}}\,\check{\chi}_\rho(x_{m'} - x_m)$. The diagonal terms ($m = m'$) give $\sum_m |b_m|^2\check{\chi}_\rho(0) = |B_\rho|\sum_m |b_m|^2$ since $\check{\chi}_\rho(0) = |B_\rho|$. For the off-diagonal part we cannot use the ball indicator's transform (a jinc/Bessel profile decaying only as $|y|^{-(d+1)/2}$), whose Schur sum diverges; instead dominate by a *smooth majorant*: choose $\varphi_0 \in C_c^\infty$ with $\varphi_0(\eta) \geq \mathbf{1}_{|\eta|\leq 1}$ and $|\check{\varphi_0}(y)| \lesssim (1 + |y|)^{-N}$ for any N (a Schwartz function since φ_0 is smooth compactly supported), set $\varphi_\rho(\xi) = \varphi_0(\xi/\rho)$, and note $\check{\varphi_\rho}(y) = \rho^d\check{\varphi_0}(\rho y)$ with $\check{\varphi_\rho}(0) = \rho^d\check{\varphi_0}(0) \asymp |B_\rho|$ and $|\check{\varphi_\rho}(y)| \lesssim \rho^d(1+\rho|y|)^{-N}$. Because $\mathbf{1}_{B_\rho} \leq \varphi_\rho$, $\int_{B_\rho}|\mathbf{a}_b|^2 d\xi \leq \int \varphi_\rho|\mathbf{a}_b|^2 d\xi = \sum_{m,m'} b_m\overline{b_{m'}}\check{\varphi}_\rho(x_{m'} - x_m)$, and the off-diagonal part is bounded by

$$\sum_{m\neq m'} |b_m||b_{m'}|\,\rho^d\big(1 + \rho|x_{m'} - x_m|\big)^{-N} \lesssim \rho^d \sum_{m,m'} \frac{|b_m||b_{m'}|}{(1 + |x_{m'} - x_m|/\delta)^N},$$

using $\rho = \Theta\delta^{-1}$ and $|x_{m'} - x_m| \geq 4\delta$. The kernel $K(x_{m'}, x_m) = (1+|x_{m'} - x_m|/\delta)^{-N}$ on the separated set is Schur-/Cotlar-summable: $\sup_{m'}\sum_m K \lesssim \sum_{k\geq 1} k^{d-1}(1+4k)^{-N} < \infty$ for $N > d$, and symmetrically in m'; hence the off-diagonal quadratic form is $\lesssim \rho^d\sum_m |b_m|^2$. Dividing by $|B_\rho| \asymp \rho^d$ gives the first bound of (7). The second (weighted) bound is the first applied to the coefficients $b_m^{(j)} = a_m x_{m,j}$ for each coordinate j: pointwise, $|\sum_m a_m(x_m\cdot\xi)e^{ix_m\cdot\xi}|^2 = |\sum_j \xi_j \sum_m a_m x_{m,j}e^{ix_m\cdot\xi}|^2 \leq |\xi|^2\sum_j|\sum_m a_m x_{m,j}e^{ix_m\cdot\xi}|^2$, and averaging each coordinate term over B_ρ with (i), using $|x_{m,j}| \leq |x_m| \leq \frac{1}{2}$ and $|\xi| \leq \rho$,

$$\frac{1}{|B_\rho|}\int_{B_\rho}\Big|\sum_m a_m(x_m\cdot\xi)e^{ix_m\cdot\xi}\Big|^2 d\xi \leq \rho^2\sum_j \mathcal{M}_\rho(a\, x_{\cdot,j}) \lesssim \rho^2\sum_j\sum_m a_m^2 x_{m,j}^2 \leq \frac{\rho^2}{4}\sum_m a_m^2 \asymp \delta^{-2}\sum_m a_m^2.$$

This proves the second bound of (7). □

Lemma 3.3 (Scale estimate for wavelet packets)**.** *Fix χ as above and $R \in (0, \frac{1}{8}]$. There are $c_1, C_1 > 0$ (depending only on d, χ, R) such that the following holds. Let $g = \sum_m \varepsilon_m a_m \psi\big(\frac{\cdot - x_m}{\delta}\big)$ be a wavelet packet built from a fixed $\psi \in C_c^\infty$ with $\operatorname{supp}\psi \subset B(0, R)$, $\psi \neq 0$, $\int \psi\, dx = 0$, where $\varepsilon_m \in \{-1, 0, 1\}$, $a_m \in \mathbb{R}$, and the centres x_m lie in the* annulus *$c_0 \leq |x_m| \leq \frac{1}{2} - c_0$ for a fixed constant $c_0 = c_0(d) \in (0, \frac{1}{16}]$ with pairwise separation $|x_m - x_{m'}| \geq 4\delta$ (so the terms have pairwise disjoint supports, all inside*

$\{|x| \le \frac{1}{2} - c_0/2\}$, *at distance* $\ge c_0/2 + R\,\delta \ge c_0/2$ *from* ∂D*; the support stays in* $\{|x| \le \frac{3}{4}\}$ *too). Then for* $\delta \in (0, \delta_0]$,

$$c_1\,\delta\,\|g\|_{L^2(D)} \;\le\; \|Tg\|_{L^2(D)} \;\le\; C_1\,\delta\,\|g\|_{L^2(D)}, \tag{8}$$

with $\|g\|_2$ *computed on* D *(equivalently* $\mathbb{R}^d$*, by the support).*

Proof. Since $\operatorname{supp} g \subset \{|x| \le \frac{3}{4}\}$ and $\chi = 1$ there, $g = g\chi$, so $\tilde{g} = g$. Write $Tg = U - q$ with $U = -\Delta_{\mathbb{R}^d}^{-1}(Lg)$ and q harmonic in D, $q = U$ on ∂D.

Whole-space term U. We use the exact identity (5): with $\Lambda F = \xi \cdot \nabla_\xi F$ the dilation generator, $U = \frac{1}{d}|D|^{-2}\Lambda L_{\mathrm{op}}\, g$ is a nonlocal Fourier operator of order exactly -1, and on a scale-δ packet it multiplies the L^2 norm by $\asymp \delta$ (the content of (6)). For concreteness and constants we compute directly. Write $\widehat{g}(\xi) = \delta^d \widehat{\psi}(\delta\xi)\mathbf{a}(\xi)$ with $\mathbf{a}(\xi) := \sum_m \varepsilon_m a_m e^{-ix_m\cdot\xi}$. Since the packet terms are disjointly supported, $\|g\|_2^2 = \sum_m a_m^2 \delta^d \|\psi\|_2^2$. With dilation generator $\Lambda F = \xi \cdot \nabla_\xi F$ applied to $F = \widehat{g}$, and splitting, $\xi \cdot \nabla_\xi \widehat{g} = \delta^d\big(B_1(\xi) + B_2(\xi)\big)$ with $B_1(\xi) = (\Lambda\widehat{\psi})(\delta\xi)\,\mathbf{a}(\xi)$, $B_2(\xi) = \widehat{\psi}(\delta\xi)\,\xi\cdot\nabla_\xi\mathbf{a}(\xi)$, and $\xi\cdot\nabla_\xi\mathbf{a}(\xi) = -i\sum_m \varepsilon_m a_m (x_m\cdot\xi)e^{-ix_m\cdot\xi}$. Cover the loaded shell $S_k = \{|\xi| \asymp \delta^{-1}\}$ by $O(1)$ balls B_ρ, $\rho \asymp \delta^{-1}$, and apply Lemma 3.2 (claims (i) and the weighted corollary): $\frac{1}{|B_\rho|}\int_{B_\rho}|\mathbf{a}|^2 \lesssim \sum a_m^2$ and $\frac{1}{|B_\rho|}\int_{B_\rho}|\xi\cdot\nabla_\xi\mathbf{a}|^2 \lesssim \rho^2\cdot\frac{1}{4}\sum a_m^2 \asymp \delta^{-2}\sum a_m^2$. Since $|\widehat{\psi}(\delta\xi)|, |(\Lambda\widehat{\psi})(\delta\xi)| \lesssim$ const, squaring and integrating gives, per contributing term,

$$\begin{aligned}
\int_{S_k} |\xi|^{-4}|\xi\cdot\nabla_\xi\widehat{g}|^2 d\xi &\lesssim \delta^{2d+4}\Big[\int_{S_k}|B_1|^2 + \int_{S_k}|B_2|^2\Big] \\
&\lesssim \delta^{2d+4}\cdot|S_k|\,\frac{1}{|B_\rho|}\Big[\int_{B_\rho}|\mathbf{a}|^2 + \int_{B_\rho}|\xi\cdot\nabla_\xi\mathbf{a}|^2\Big] \\
&\lesssim \delta^{2d+4}\cdot\delta^{-d}\Big(\sum a_m^2 + \delta^{-2}\sum a_m^2\Big) \;\lesssim\; \delta^2\cdot\delta^d\sum a_m^2 \;=\; C\,\delta^2\,\frac{\|g\|_2^2}{\|\psi\|_2^2}.
\end{aligned} \tag{9}$$

Here the power count is: δ^{2d} (prefactor) $\times\delta^4$ ($|\xi|^{-4}$) $\times\delta^{-d}$ (shell volume) $\times\delta^{-2}$ (weighted AO, dominant B_2 term) $= \delta^2$; the B_1 term is one power of δ smaller. The constant is uniform in M since there are $O(1)$ balls.

For the off-shell shells S_j: those with $2^j \gg \delta^{-1}$ use the Schwartz decay $|\widehat{\psi}(\delta\xi)| \lesssim (\delta|\xi|)^{-N}$ so each contributes $\delta^2\cdot 2^{-N(j-k)}$; those with $2^j \ll \delta^{-1}$ use the mean-zero bound $|\widehat{\psi}(\delta\xi)| \lesssim \delta|\xi|$ ($\int\psi = 0$), contributing $\delta^2\cdot 2^{-2(k-j)}$. Both tails sum geometrically, so the whole-space term satisfies the upper bound

$$\|U\|_{L^2(\mathbb{R}^d)}^2 \;\lesssim\; \delta^2\|g\|_2^2, \qquad \text{uniformly in } M, \delta. \tag{10}$$

Lower bound (loaded-shell diagonal dominance). Retain only the loaded shell $S_k = \{|\xi| \asymp \delta^{-1}\}$ (all terms non-negative on the Fourier side). Since $\|U\|_2^2 \ge \int_{S_k}|\xi|^{-4}|\widehat{Lg}|^2 d\xi \asymp \delta^4\int_{S_k}|\widehat{Lg}|^2 d\xi$ and by (5) $\widehat{Lg} = -\frac{1}{d}\xi\cdot\nabla_\xi\widehat{g}$, using $|a+b|^2 \ge \frac{1}{2}|b|^2 - |a|^2$ with $b = \delta^d\widehat{\psi}(\delta\xi)\xi\cdot\nabla_\xi\mathbf{a}(\xi)$ and $a = \delta^d(\Lambda\widehat{\psi})(\delta\xi)\mathbf{a}(\xi)$:

$$\|U\|_2^2 \ge \frac{\delta^4}{2d^2}\int_{S_k}\delta^{2d}|\widehat{\psi}(\delta\xi)|^2|\xi\cdot\nabla_\xi\mathbf{a}(\xi)|^2 d\xi - \frac{\delta^4}{d^2}\int_{S_k}\delta^{2d}|(\Lambda\widehat{\psi})(\delta\xi)|^2|\mathbf{a}(\xi)|^2 d\xi.$$

Choose once and for all the base profile ψ of the lemma statement: a radial $\psi \in C_c^\infty$ with $\operatorname{supp}\psi \subset B(0,R)$, $\int\psi = \widehat{\psi}(0) = 0$, and $\widehat{\psi}$ nonvanishing on the closed annulus $\{1 \le |\eta| \le 2\}$ (readily constructed by applying a power of $-\Delta$ to a radial C_c^∞-bump and adding a small correction). Thus $|\widehat{\psi}(\delta\xi)| \ge c_\psi > 0$ on the loaded shell while $|(\Lambda\widehat{\psi})(\delta\xi)| \le C_\psi$. Now $\xi\cdot\nabla_\xi\mathbf{a}(\xi) = -i\sum_m \varepsilon_m a_m (x_m\cdot\xi)e^{-ix_m\cdot\delta\xi}$; because the centres lie in the annulus $|x_m| \ge c_0$, on the shell $|x_m\cdot\xi| \asymp |\xi||x_m| \ge \frac{c_0}{2}\delta^{-1}$ (in the diagonal/phase-gradient part). Apply Lemma 3.2's weighted corollary with the *lower* direction: the oscillatory off-diagonals are Schur-summable (by the smooth majorant bound of the lemma) and are dominated by the diagonal, so

$$\int_{S_k}|\widehat{\psi}(\delta\xi)|^2|\xi\cdot\nabla_\xi\mathbf{a}(\xi)|^2 d\xi \;\ge\; c_\psi^2\,\delta^{-d}\,\delta^{-2}\,\frac{c_0^2}{4}\sum_m a_m^2 \;\big(\text{diagonal } m = m'\big),$$

the $m = m'$ terms giving $\sum_m a_m^2 \int_{S_k} |x_m \cdot \xi|^2 d\xi \asymp \sum_m a_m^2 \delta^{-d}\delta^{-2}$ since $\int_{S_k} |x_m \cdot \xi|^2 d\xi \asymp |x_m|^2 \delta^{-2-d} \cdot \delta^2 = |x_m|^2\delta^{-d} \gtrsim c_0^2 \delta^{-d}$. Similarly the second (subtracted) term is bounded above by the *unweighted* AO: $\le C_\psi \delta^{-d} \sum a_m^2$ (no δ^{-2}). Hence

$$\|Ug\|_2^2 \;\ge\; \delta^{4+2d}\,\delta^{-d}\Big(\frac{c_\psi^2 c_0^2}{8d^2}\,\delta^{-2}\sum a_m^2 - \frac{C_\psi}{d^2}\sum a_m^2\Big) \;\ge\; c\,\delta^{d+4}\cdot\delta^{-2}\sum a_m^2 \;=\; c\,\delta^2\cdot\delta^d\sum a_m^2 \;=\; c\,\delta^2\,\frac{\|g\|_2^2}{\|\psi\|_2^2},$$

using $\delta^{4+2d}\delta^{-d}\delta^{-2} = \delta^{d+2} = \delta^2 \cdot \delta^d$, and for $\delta \le \delta_0(c_\psi, c_0, C_\psi)$ the dominant δ^{-2} term controls the subtracted one. Hence $\|U\|_2 \ge c'_U\,\delta\|g\|_2$ with $c'_U = c'_U(d, \psi, c_0) > 0$.

Harmonic correction q. By Lemma 2.5 (applied with $F = Lg = \frac{1}{d}\operatorname{div}(xg)$, for which the solution $w = Tg$ splits as $w = U - q$ with exactly the present U and q), $U = (-\Delta_{\mathbb{R}^d})^{-1}F$ is a convolution with a locally-integrable kernel of a field supported in $D_{(\tau_0)}$, $\tau_0 = \frac{1}{8}$, and on ∂D one has the interior bound $\|U\|_{L^\infty(\partial D)} \lesssim \|g\|_2$ (Cauchy–Schwarz in y, distance $\ge \tau_0/4$ to the support). Split $q = q_0 + q_{\mathrm{osc}}$ with $q_0 = |\partial D|^{-1}\int_{\partial D} q$ the constant Poisson mode; the maximum principle gives $\|q_0\|_{L^2} \le |\partial D|^{-1}\int_{\partial D}|U| \lesssim \delta\|g\|_2$ by the bump-wise multipole expansion of $U|_{\partial D}$ (monopole killed by $\int\psi = 0$, first-order remainder $O(\delta\|\psi_m\|_1)$, $\sum_m |a_m|\|\psi_m\|_1 \lesssim \|g\|_2$), and the mean-zero part decays into the interior at the Poisson rate: $q_{\mathrm{osc}} = \sum_\Lambda \rho^{\sqrt{\Lambda}}\widehat{U}(\Lambda)$, so on $\{|x| \le \frac{3}{4}\}$ (which carries the shell mass of U) $\|q_{\mathrm{osc}}\|_{L^2} \le \big(\max_\Lambda \rho^{2\sqrt{\Lambda}}\big)^{1/2}\|U\|_{L^2(\partial D)} \lesssim e^{-c/\delta}\|g\|_2 \ll \delta\|g\|_2$ (the largest $\sqrt{\Lambda}$ is $\asymp \delta^{-1}$, and $\rho^{2\sqrt{\Lambda}} \le (\frac{3}{4})^{2c\delta^{-1}} \le e^{-c'/\delta}$). Hence

$$\|q\|_{L^2(|x|\le\frac{3}{4})} \;\lesssim\; \delta\,\|g\|_2, \qquad \text{constant depending on } (d, \psi, c_0, \tau_0), \tag{11}$$

$\{|x| \le \frac{3}{4}\}$ being where the shell mass of U lies ($\operatorname{supp} g \subset \{|x| \le \frac{3}{4}\}$, the $\mathbb{R}^d$ integral dominated by its $\{|x| \le \frac{3}{4}\}$ part up to the $e^{-c/\delta}$ tail already accounted).

Combining, $\|Tg\|_2 \le \|U\|_2 + \|q\|_2 \lesssim \delta\|g\|_2$ and $\|Tg\|_2 \ge \|U\|_{L^2(|x|\le\frac{3}{4})} - \|q\|_{L^2(|x|\le\frac{3}{4})} \ge (c'_U/2 - C_q\delta)\,\delta\|g\|_2$ for the $\{|x| \le \frac{3}{4}\}$-restricted shell lower bound $c'_U/2$ (same argument, all estimates on the shrunk domain) and C_q the constant of (11); with δ_0 shrunk so $C_q\delta_0 \le c'_U/4$,

$$\|Tg\|_2 \;\ge\; \tfrac{c'_U}{4}\,\delta\|g\|_2 =: c_1\delta\|g\|_2, \qquad \|Tg\|_2 \;\le\; (C_U + C_q)\delta\|g\|_2 =: C_1\delta\|g\|_2,$$

with c_1, C_1, δ_0 depending only on (d, ψ, c_0). (The origin-centre modes, excluded here by the annulus hypothesis, are treated in Lemma 5.2 via the exact radial computation.) □

Remark 3.4 (Constant bookkeeping in (8)). Both $C''_E = C''_E(d)$ (trace/elliptic constant of the unit ball) and $c'_U = c'_U(d, \psi, c_0)$ are (α, n, δ, r)-independent, so the threshold $\delta_0 = \delta_0(c_\psi, c_0, C_\psi, C''_E)$ and all constants below keep the (α, d, ψ)-dependence only; boundary-near bumps are *not* asked to be small, the loss being absorbed by shrinking δ_0.

Remark 3.5 (T is a degree-one operator, globally). For a full wavelet expansion $h = \sum_{j \ge j_0} h_j$ with h_j at scale 2^{-j}, applying Lemma 3.3 to each dyadic piece and summing (the pieces are almost orthogonal in L^2, and T maps the scale 2^{-j} piece into functions still almost-orthogonal up to the operator's pseudo-local tail) yields $\|Th\|_2^2 \lesssim \sum_j 2^{-2j}\|h_j\|_2^2 \lesssim \|h\|_{H^{-1}}^2$, i.e. T maps H^s into H^{s+1} type spaces with exactly one derivative of smoothing, globally on the Besov scale. In particular $\|Th\|_2 \le C\|h\|_2$, the only global fact (besides (8)) used below.

4 Lower bound

Proposition 4.1. *There exist $r_0 = r_0(\alpha, d) \in (0, 1]$ and $c > 0$ such that for all $r \in (0, r_0]$ and all large n,*

$$\inf_{\widehat{f}_n}\sup_{f_0 \in \mathcal{F}(\alpha, r)} \mathbb{E}_{f_0}\|\widehat{f}_n - f_0\|_2^2 \;\ge\; c\,n^{-\frac{2\alpha}{2\alpha+d+2}}.$$

Proof. **Step 1 (wavelet packet).** Fix the single base profile $\psi \in C_c^\infty(\mathbb{R}^d)$ of Lemma 3.3 (radial, $\operatorname{supp}\psi \subset \{|x| \le \frac{1}{4}\}$, $\int \psi = \widehat{\psi}(0) = 0$, $\widehat{\psi}$ nonvanishing on $\{1 \le |\eta| \le 2\}$), normalized so $\|\psi\|_{C^\alpha} = 1$; then $\|\psi\|_2 > 0$ and $\|\psi\|_{C^\alpha} = 1$ are fixed (α, d)-constants. For $\delta \in (0, \delta_0]$ choose a maximal collection $\{x_m\}_{m=1}^M \subset \{c_0 \le |x| \le \frac{1}{2}\}$ with pairwise distance $\ge 4\delta$ (hence $M \asymp \delta^{-d}$), and set $\psi_m(x) = \psi\big(\frac{x - x_m}{\delta}\big)$. Then $\operatorname{supp}\psi_m \subset B(x_m, \delta/4)$, $\|\psi_m\|_\infty = \|\psi\|_\infty$, $\|\psi_m\|_2 = \delta^{d/2}\|\psi\|_2$, $\|\psi_m\|_{C^\alpha} = \delta^{-\alpha}\|\psi\|_{C^\alpha}$ (the C^α norm is $\asymp \delta^{-\alpha}$ for $\delta \le 1$: derivatives scale as δ^{-j} and the top t-Hölder/Zygmund seminorm as $\delta^{-\alpha}$, dominating the lower-order terms). The supports of the ψ_m are pairwise disjoint and contained in $\{|x| \le \frac{3}{4}\}$, so every partial sum $\sum_{m \in S} \pm\psi_m$ is a wavelet packet of the form treated in Lemma 3.3 (fixed base profile ψ, centres in $\{c_0 \le |x| \le \frac{1}{2}\}$, separation $\ge 4\delta$; the annulus restriction keeps the two-sided scale estimate of Lemma 3.3).

Step 2 (hypotheses). By Varshamov–Gilbert, there is a family $\mathcal{S}$ of subsets $S \subset \{1, \dots, M\}$ with $|S| = M/2$, each pair $S \ne S'$ at Hamming distance $|S \triangle S'| \ge M/8$, and $\log|\mathcal{S}| \ge c_0 M$ for an absolute $c_0 > 0$. Index hypotheses by $S \in \mathcal{S}$ and a sign/scale parameter $\kappa > 0$ (fixed below):

$$f_S = \mathbf{1} + \kappa\delta^\alpha \Psi_S, \qquad \Psi_S := \sum_{m \in S} \psi_m.$$

Because the ψ_m have disjoint supports with the same profile, $\|\Psi_S\|_{C^\alpha} \lesssim \max_m \|\psi_m\|_{C^\alpha} \asymp \delta^{-\alpha}\|\psi\|_{C^\alpha}$: the C^a norm of the sum is the max over the summands (disjoint supports), and the top t-Hölder/Zygmund difference quotient satisfies $\sup_{x,y} |\nabla^a \Psi_S(x) - \nabla^a \Psi_S(y)|/|x-y|^t \le 2^{1-t}\delta^{-\alpha}\|\nabla^a\psi\|_\infty + \delta^{-\alpha}[\psi]_{\dot{C}^t}$ — cross-support quotients are the same order as single-bump ones, giving comparability with a ψ-dependent constant $\le 2^{1-t} + 1$. Hence

$$\|f_S - \mathbf{1}\|_{C^\alpha} \lesssim \kappa\delta^\alpha \cdot \delta^{-\alpha}\|\psi\|_{C^\alpha} = \kappa\|\psi\|_{C^\alpha} \;\le\; r$$

provided $\kappa \le r\|\psi\|_{C^\alpha}^{-1}$, i.e. $\kappa \le r$ since $\|\psi\|_{C^\alpha} = 1$ (Step 1). We take the maximal admissible $\kappa = r$; the separation below is then $\varepsilon^2 \asymp r^2\delta^{2\alpha}$, so the lower-bound constant inherits the factor r^2 of the shrinking class $\mathcal{F}(\alpha, r)$ – a fixed positive constant for every fixed $r \in (0, r_0]$. Thus $f_S \in \mathcal{F}(\alpha, r)$ for all $S \in \mathcal{S}$. Also, by disjoint supports,

$$\|\Psi_S\|_2^2 = |S|\,\|\psi_m\|_2^2 \asymp M\,\delta^d \asymp 1, \qquad \|\Psi_S\|_\infty = \|\psi\|_\infty.$$

Step 3 (separation). For $S \ne S'$, $f_S - f_{S'} = \kappa\delta^\alpha \sum_{m \in S \triangle S'} \pm\psi_m$, the supports disjoint, so

$$\|f_S - f_{S'}\|_2^2 = \kappa^2\delta^{2\alpha}|S \triangle S'|\,\|\psi_m\|_2^2 \asymp \kappa^2\delta^{2\alpha} \cdot M \cdot \delta^d \asymp \kappa^2\delta^{2\alpha}.$$

So the family is ε-separated with $\varepsilon^2 \asymp \kappa^2\delta^{2\alpha}$.

Step 4 (KL control). With $g_S := f_S - \mathbf{1} = \kappa\delta^\alpha\Psi_S$: g_S is supported in $\{|x| \le \frac{3}{4}\}$, is a single-scale packet of the form in Lemma 3.3 (fixed profile ψ with $\int\psi = 0$, centres in $\{|x| \le \frac{1}{2}\}$ separated by $\ge 4\delta$; each ψ_m has $\|\nabla\psi_m\|_2 \asymp \delta^{-1}\|\psi_m\|_2$), so by Lemma 3.3,

$$\|Tg_S\|_2 \;\asymp\; \delta\|g_S\|_2 \;\asymp\; \delta \cdot \kappa\delta^\alpha\|\Psi_S\|_2 \;\asymp\; \kappa\delta^{\alpha+1}.$$

(Here $\|\Psi_S\|_2 \asymp 1$ by Step 2.) The quadratic remainder, by Lemma 2.7,

$$\|R(g_S)\|_2 \lesssim \|g_S\|_\infty\|g_S\|_2 = \kappa\delta^\alpha\|\psi\|_\infty \cdot \kappa\delta^\alpha\|\Psi_S\|_2 \asymp \kappa^2\delta^{2\alpha}.$$

Since $\alpha > 1$, $\|R(g_S)\|_2 \lesssim \kappa^2\delta^{2\alpha} \ll \kappa\delta^{\alpha+1} \asymp \|Tg_S\|_2$ for δ small (the ratio is $\kappa\delta^{\alpha-1} \to 0$ — the decisive use of $\alpha > 1$). Therefore

$$\|u_{f_S} - u_0\|_2 \le \|Tg_S\|_2 + \|R(g_S)\|_2 \lesssim \kappa\delta^{\alpha+1},$$

and by (1),

$$\mathrm{KL}(P_{f_S} \,\|\, P_{f_0}) = \frac{n}{2}\|u_{f_S} - u_0\|_2^2 \le Cn\kappa^2\delta^{2\alpha+2}. \tag{12}$$

Step 5 (Fano). Take $\delta = c\, n^{-1/(2\alpha+d+2)}$. Then the balance is

$$n\delta^{2\alpha+2} = c^{2\alpha+2} n^{1-\frac{2\alpha+2}{2\alpha+d+2}} = c^{2\alpha+2} n^{\frac{d}{2\alpha+d+2}} = c^{2\alpha+2}\delta^{-d}.$$

Choosing $c > 0$ small makes $Cn\kappa^2\delta^{2\alpha+2} \le \frac{1}{10}\,\delta^{-d} \le \frac{1}{10}\cdot\frac{\log|\mathcal{S}|}{c_0}$, i.e. the average KL is $\le \frac{1}{10}\log|\mathcal{S}|$. With $\varepsilon^2 \asymp \kappa^2\delta^{2\alpha} = \kappa^2 c^{2\alpha} n^{-\frac{2\alpha}{2\alpha+d+2}}$ and $\log|\mathcal{S}| \ge c_0 M \asymp \delta^{-d}$, Fano's inequality (Lemma 4.2) gives

$$\max_{S\in\mathcal{S}} \mathbb{E}_{f_S}\|\widehat{f} - f_S\|_2^2 \;\ge\; \frac{\varepsilon^2}{8}\Big(1 - \frac{1}{10} - \frac{\log 2}{\log|\mathcal{S}|}\Big) \;\ge\; c'\, n^{-\frac{2\alpha}{2\alpha+d+2}}$$

for n large. Since $\{f_S\} \subset \mathcal{F}(\alpha, r)$, the minimax risk over $\mathcal{F}(\alpha, r)$ is at least this, proving Proposition 4.1. □

Lemma 4.2 (Fano, applied form)**.** *Let $|\Theta| = N \ge 2$, $\{P_\theta\}_{\theta\in\Theta}$ probability measures with $\frac{1}{N}\sum_\theta \mathrm{KL}(P_\theta\|P_0) \le \beta\log N$, $\beta < 1$, and parameter values separated: $\|f_\theta - f_{\theta'}\|_2 \ge \varepsilon$ for $\theta \ne \theta'$. Then*

$$\inf_{\widehat{f}}\max_\theta \mathbb{E}_\theta\|\widehat{f} - f_\theta\|_2^2 \;\ge\; \frac{\varepsilon^2}{4}\Big(1 - \beta - \frac{\log 2}{\log N}\Big)\cdot\frac{1}{2} \;\ge\; \frac{\varepsilon^2}{8}\Big(1 - \beta - \frac{\log 2}{\log N}\Big).$$

Proof. Given any estimator $\widehat{f}$, let $\widehat{\theta} = \arg\min_{\theta\in\Theta}\|\widehat{f} - f_\theta\|_2$. By ε-separation and the triangle inequality, $\|\widehat{f} - f_\theta\|_2 \ge \varepsilon/2$ whenever $\widehat{\theta} \ne \theta$. Markov gives $\mathbb{E}_\theta\|\widehat{f} - f_\theta\|_2^2 \ge \frac{\varepsilon^2}{4}P_\theta(\widehat{\theta} \ne \theta)$. Averaging and using $\max_\theta \ge \frac{1}{N}\sum_\theta$: $\max_\theta \mathbb{E}_\theta\|\widehat{f} - f_\theta\|_2^2 \ge \frac{\varepsilon^2}{4}\cdot\frac{1}{N}\sum_\theta P_\theta(\widehat{\theta} \ne \theta)$. Fano's inequality bounds the average testing error below by $1 - \frac{\beta\log N + \log 2}{\log N} = \big(1 - \beta - \frac{\log 2}{\log N}\big)$. Combining gives the claim (Tsybakov, *Introduction to Nonparametric Estimation*, Theorem 2.5). □

5 Upper bound

Proposition 5.1. *There is an estimator $\widehat{f}_n$ with*

$$\sup_{f_0\in\mathcal{F}(\alpha,r)} \mathbb{E}_{f_0}\|\widehat{f}_n - f_0\|_2^2 \;\le\; C\, n^{-\frac{2\alpha}{2\alpha+d+2}}.$$

The construction: (A) discretize the observation and set up a scale-δ sieve least-squares problem; (B) a discrete inf–sup/smoothing inequality inverting T on the sieve; (C) balance bias, variance, and the quadratic remainder.

5.1 (A) Observation discretization; sieve least squares

Let $V_\delta \subset L^2(D)$ be the span of the Dirichlet eigenfunctions of $-\Delta$ with eigenvalue $\le C_W\,\delta^{-2}$ (C_W a fixed large constant chosen in Lemma 5.2), i.e. the eigenfunctions oscillating at frequencies $\lesssim \delta^{-1}$; by Weyl's law $\dim V_\delta \asymp \delta^{-d}$ (wavelength-2^{-j} eigenfunctions carry eigenvalues $\asymp 2^{2j}$, so the cutoff $\lesssim \delta^{-2}$ keeps *physical scale* $\gtrsim \delta$). Let P_δ be the orthogonal projection onto V_δ and set

$$Y_\delta := P_\delta Y = P_\delta u_{f_0} + n^{-1/2}\xi_\delta, \qquad \xi_\delta := P_\delta\mathbb{W}, \quad \mathbb{E}\|\xi_\delta\|_2^2 = \dim V_\delta \asymp \delta^{-d}.$$

The sieve must avoid both degeneracies of the transport field $\nabla u_0 = x/d$: the origin (where it vanishes) and the boundary collar (where the Hölder boundary estimate loses one degree). We therefore estimate only interior content. Let $\mathcal{F}_\delta$ be the span of all wavelet packets at scales $\ge \delta$ *whose centres lie in the fixed interior* $D_1 := \{c_1 \le |x| \le \frac{1}{2}\}$ for a small fixed $c_1 = c_1(d) > 0$, intersected with $\mathcal{F}(\alpha, r)$. Fix a patch ζ, $\equiv 1$ on $\mathrm{supp}(\mathcal{F}_\delta - \mathbf{1})$, $\equiv 0$ on $B_{c_1/2} \cup \{|x| > \frac{3}{4}\}$. For a general $f \in \mathcal{F}(\alpha, r)$ write $f = f_{\le\delta} + f_{>\delta}$ (scales $\ge \delta$ and $< \delta$; $\mathrm{supp}(f_{\le\delta} - \mathbf{1}) \subset \{\zeta = 1\}$) and set

$$\Pi_\delta f \;:=\; \mathbf{1} + \big(f_{\le\delta} - \mathbf{1}\big)\,\zeta, \qquad \text{so } \Pi_\delta f = \mathbf{1} \text{ off } \mathrm{supp}\,\zeta \text{ (a ball about 0 and a collar of } \partial D). \tag{13}$$

Thus $\Pi_\delta f \in \mathcal{F}(\alpha, 2r)$, and the comparison point $m = \Pi_\delta f_0$ is Dirichlet ($u_m = 0$ on ∂D, as is u_{f_0}, by (4)) and interior-regular. By the standard $B^\alpha_{\infty,\infty} = C^\alpha$ wavelet characterization,

$$\|f_{>\delta}\|_\infty \le C\delta^\alpha, \quad \|f_{>\delta}\|_2 \le C\delta^\alpha, \quad \|f_{\le\delta}\|_{C^\alpha} \le C\|f\|_{C^\alpha}, \quad \dim \mathcal{F}_\delta \lesssim \delta^{-d}. \tag{14}$$

The *bias* $e = \Pi_\delta f - f = -f_{>\delta} + (f_{\le\delta} - \mathbf{1})(\zeta - \mathbf{1})$ has $\|e\|_2 \lesssim \delta^\alpha$ by (14); its dominant content is interior (the origin/boundary remainder is handled in the collar split of (23)). The projection P_δ retains the part of u_{f_0} oscillating at scales $\gtrsim \delta$ and makes the noise finite-dimensional, so the estimator and its risk are well-defined on the white-noise model. Define the sieve least-squares estimator

$$\widehat{f} = \arg\min_{g \in \mathcal{F}_\delta} \|P_\delta u_g - Y_\delta\|_2^2.$$

Since $\mathcal{F}_\delta$ is closed and $g \mapsto \|P_\delta u_g - Y_\delta\|_2$ is continuous and coercive ($\|P_\delta u_g\|_2 \to \infty$ with $\|g\|_2$ on the sieve by (15) and the apriori bound, Lemma 2.6), a measurable minimizer $\widehat{f}$ exists (selection theorem on the closed finite-dimensional sieve); only its risk is analyzed below.

5.2 (B) Discrete inf–sup (scale inversion on the sieve)

Lemma 5.2. *There are $c, C > 0$ such that for all small δ the following holds. Let $h = \sum_\lambda h_\lambda$ be a wavelet-packet expansion where each h_λ is a packet of Lemma 3.3-type with centre x_λ and scale $\delta_\lambda \ge \delta$ (on the sieve of §(A), every packet has its centre in the interior annulus, so the subfamily below* is *the whole sieve error). For the subfamily with centres in the annulus $c_0 \le |x_\lambda| \le \frac{1}{2}$ (notation $h^{\mathrm{bulk}} = \sum_{\lambda:\, c_0 \le |x_\lambda| \le \frac{1}{2}} h_\lambda$),*

$$c\,\delta\|h^{\mathrm{bulk}}\|_2 \;\le\; \|P_\delta T h^{\mathrm{bulk}}\|_2 \;\le\; C\,\delta\|h^{\mathrm{bulk}}\|_2, \qquad \textit{and more globally} \qquad \|Th\|_2 \le C\|h\|_2. \tag{15}$$

For origin-near modes ($|x_\lambda| \le \rho\delta_\lambda$ for a fixed ρ) the transport field degenerates and the gain is only $\gtrsim \delta_\lambda^2$ (Lemma 5.3); they are excluded *from the sieve of §(A) and projected into the bias, see Remark 5.4.*

Lemma 5.3 (Exact radial gain near the origin)**.** *Let $g(x) = \varphi(|x|/\delta)$ be radial, supported in B_δ about the origin, $\|g\|_2 \asymp \delta^{d/2}$. Then the solution $w = Tg$ of the radial part of (3) satisfies $w(x) = -\frac{1}{d}\int_{|x|}^\delta s\, g(s)\, ds$ for $|x| \le \delta$, $w \equiv 0$ for $|x| \ge \delta$, and $\|Tg\|_2 \asymp \delta^2\|g\|_2$. More generally, for a mode centred at distance $|x_\lambda|$ with scale δ_λ, the mode-wise gain is at least the radial endpoint: for an origin-near mode ($|x_\lambda| \le \rho\,\delta_\lambda$, ρ fixed),*

$$\|Th_\lambda\|_2 \;\gtrsim\; \delta_\lambda^2\,\|h_\lambda\|_2. \tag{16}$$

(used in Remark 5.4 to record the degeneracy).

Proof of Lemma 5.3. For the radial bump, (3) reads $\Delta w = Lg = \frac{1}{d}\operatorname{div}(xg)$, i.e. $\frac{1}{r^{d-1}}\partial_r(r^{d-1}w') = \frac{1}{d}\frac{1}{r^{d-1}}\partial_r(r^d g)$, so $r^{d-1}w'(r) = \frac{1}{d}r^d g(r) + C$; regularity at $r = 0$ forces $C = 0$, so $w'(r) = \frac{r}{d}g(r)$ for all $r > 0$. On $[\delta, 1]$ the source $g \equiv 0$, hence $w' = 0$ there, and the boundary condition $w(1) = 0$ gives $w \equiv 0$ on $[\delta, 1]$. Continuity of w at $r = \delta$ (the flux w' is bounded) then yields $w(x) = -\frac{1}{d}\int_{|x|}^\delta s\, g(s)\, ds$ for $|x| \le \delta$, and

$$\|w\|_2^2 = |\mathbb{S}^{d-1}| \int_0^\delta \Big|\frac{1}{d}\int_r^\delta s g(s)\, ds\Big|^2 r^{d-1} dr \;\asymp\; \delta^4\|g\|_2^2$$

by Cauchy–Schwarz (both sides $\asymp \delta^4\|g\|_2^2$, e.g. the lower bound from profiles with mass near $s = \delta$ and the upper from $|\int_r^\delta sg| \le \delta\|g\|_{L^2(r,\delta)}$ and the δ-length). Thus $\|Tg\|_2 \asymp \delta^2\|g\|_2$.

For the one-sided bound (16), re-run Lemma 3.3's loaded-shell power count; the *only* change is the shift factor $|x_m|^2$ in the weighted almost-orthogonality. With that proof's notation ($\delta = \delta_\lambda$; shell $S_k = \{|\xi| \asymp \delta_\lambda^{-1}\}$, volume $\asymp \delta_\lambda^{-d}$; $\rho_k \asymp \delta_\lambda^{-1}$), the shell integrand $|\xi|^{-4}|\widehat{Lg}|^2$ has its dominant $\xi \cdot \nabla_\xi \mathbf{a}$

factor bounded by $|S_k|\rho_k^2\sum_m|x_m|^2a_m^2$ (Lemma 3.2, weighted part). Annulus centres have $|x_m|\asymp 1$; origin-near centres have $|x_m|\lesssim\delta_\lambda$, so $|x_m|^2\lesssim\delta_\lambda^2$ contributes one full extra power of δ_λ: the count is

$$\|U\|_2^2 \;\gtrsim\; \underbrace{\delta_\lambda^4}_{|\xi|^{-4}}\;\underbrace{\delta_\lambda^{2d}}_{\text{prefactor}}\;\underbrace{\delta_\lambda^{-d}}_{|S_k|}\;\underbrace{\rho_k^2}_{\asymp\delta_\lambda^{-2}}\;\underbrace{\delta_\lambda^2}_{|x_m|^2}\sum_m a_m^2 \;\asymp\; \delta_\lambda^4\,(\delta_\lambda^d\sum_m a_m^2) \;=\; \delta_\lambda^4\|h_\lambda\|_2^2,$$

using $\|h_\lambda\|_2^2\asymp\delta_\lambda^d\sum a_m^2$. The subtracted $(\Lambda\widehat{\psi})$-term lacks the ρ_k^2 amplification and is one power smaller, as in Lemma 3.3; the harmonic correction $q=O(\delta_\lambda\|h_\lambda\|_2)$ by (11) absorbs into δ_0. Hence $\|Th_\lambda\|_2\gtrsim\delta_\lambda^2\|h_\lambda\|_2$ (constants depending on (d,ψ,c_0,ρ) only). □

Proof of Lemma 5.2. We prove the two-sided estimate for the bulk subfamily (the whole sieve error on the interior sieve of §(A)) and the global bound; the origin modes it excludes are projected into Π_δ (Remark 5.4). Throughout, P_δ is the spectral projection of the Dirichlet Laplacian onto $\lambda_\nu\le C_W\,\delta^{-2}$ (spatial frequencies $\lesssim\delta^{-1}$, matching the one-degree smoothing; see §(A)).

(i) Lower bound on $\|Th^{\mathrm{bulk}}\|_2$. Decompose $h^{\mathrm{bulk}}=\sum_j h_j$ into wavelet pieces at scales $2^{-j}\ge\delta$ with centres in the annulus (each h_j satisfies the hypotheses of Lemma 3.3 with scale $\delta_j=2^{-j}$ and annulus centres). By Lemma 3.3's lower bound applied per scale, $\|Th_j\|_2\gtrsim 2^{-j}\|h_j\|_2$, and the cross-terms $\langle Th_j,Th_{j'}\rangle$ for $j\ne j'$ are almost-orthogonal by Lemma 3.2 (each Th_j remains concentrated near scale 2^{-j}, both by the Fourier localization of the source and by the Schur-summable interference off the diagonal); hence

$$\|Th^{\mathrm{bulk}}\|_2^2 \;\gtrsim\; \sum_j 2^{-2j}\|h_j\|_2^2 \;\ge\; \delta^2\sum_j\|h_j\|_2^2 \;\asymp\; \delta^2\|h^{\mathrm{bulk}}\|_2^2,$$

the last equivalence by the square-function characterization (T5). Hence $\|Th^{\mathrm{bulk}}\|_2\ge c\delta\|h^{\mathrm{bulk}}\|_2$.

(ii) Spectral tail $\|(I-P_\delta)Th\|_2$. The projector cuts at $|\xi|\asymp\sqrt{C_W}\,\delta^{-1}$, retaining scales $\gtrsim\delta$; the mass of Th^{bulk} beyond this is the off-shell tail of Lemma 3.3. For a single scale-δ packet, (10)'s off-shell shells S_j, $2^j\gg\delta^{-1}$, contribute a factor $2^{-N(j-k)}$ each via the Schwartz decay $|\widehat{\psi}(\delta\xi)|\lesssim(\delta|\xi|)^{-N}$ (any fixed N), so on $\{|\xi|\gtrsim\sqrt{C_W}\delta^{-1}\}$ the mass is

$$\|P_{\{|\xi|\gtrsim\sqrt{C_W}\delta^{-1}\}}\,U\|_2^2 \;\lesssim\; \sum_{j:\,2^j\gtrsim\sqrt{C_W}\delta^{-1}} 2^{-N(j-k)}\,\delta^2\|g\|_2^2 \;\lesssim\; C_W^{-N/2}\,\delta^2\|g\|_2^2,$$

(geometric sum over off-shell shells, $2^{j-k}\gtrsim\sqrt{C_W}$). The harmonic correction q is smooth into any Sobolev norm with constant $O(\delta\|g\|_2)$ by (11), so its high tail is Schwartz-small and absorbs into the same bound. For coarser pieces $2^{-j}\in[\delta,1]$ the cutoff $\sqrt{C_W}\delta^{-1}$ sits even deeper in the off-shell tail, so the bound is uniform; summing over scales as in step (i),

$$\|(I-P_\delta)Th^{\mathrm{bulk}}\|_2 \;\le\; \widetilde{C}\,C_W^{-N/4}\,\delta\|h^{\mathrm{bulk}}\|_2, \tag{17}$$

with $\widetilde{C}=\widetilde{C}(d,\psi,c_0)$. Choosing the fixed constant C_W (hence the sieve V_δ) so large that $\widetilde{C}\,C_W^{-N/4}\le c/4$ ($N\ge4$ fixed, C_W independent of δ) makes the tail $\le\frac{c}{4}\delta\|h^{\mathrm{bulk}}\|_2$.

(iii) Combination. By the triangle inequality (ordinary norm subtraction),

$$\|P_\delta Th^{\mathrm{bulk}}\|_2 \;\ge\; \|Th^{\mathrm{bulk}}\|_2-\|(I-P_\delta)Th^{\mathrm{bulk}}\|_2 \;\ge\; c\,\delta\|h^{\mathrm{bulk}}\|_2-\tfrac{c}{4}\,\delta\|h^{\mathrm{bulk}}\|_2 \;=\; \tfrac{3c}{4}\,\delta\|h^{\mathrm{bulk}}\|_2,$$

combining step (i) and (17); both terms are $O(\delta\|\cdot\|_2)$ and the tail constant $c/4<c$, so the difference is positive. The upper bound $\|P_\delta Th^{\mathrm{bulk}}\|_2\le\|Th^{\mathrm{bulk}}\|_2\lesssim\delta\|h^{\mathrm{bulk}}\|_2$ and the global $\|Th\|_2\le C\|h\|_2$ follow from Lemma 3.3 / Remark 3.5. □

Remark 5.4 (Origin content and the interior sieve). Lemma 5.3 records that the transport field's degeneracy makes the origin the *least* revealing region (gain only $\asymp \delta_\lambda^2$ there, versus $\asymp \delta_\lambda$ in the bulk annulus). Since an adversarial h could concentrate at the origin — where inversion against the $O(\delta)$ noise/bias terms would lose a factor δ^{-1} — the sieve $\mathcal{F}_\delta$ is restricted to bulk-annulus packets, and the origin/boundary content is projected into $\Pi_\delta f_0$ (hence into the bias, whose u-image is handled by the degree-one gain of (23)). Accordingly, in §(C) the lower bound of (15) is applied to the *whole* sieve error h, supported in the bulk annulus by construction, and the analysis uses the annulus two-sided estimate of Lemma 3.3, valid for all $\alpha > 1$, $d \geq 2$ (Lemmas 3.3; c_0, c_ψ, C_E'' absorbed).

5.3 (C) Risk bound

Let $\delta = \delta_*$ at the end; for now $\delta > 0$ general. Decompose, with the comparison point $m := \Pi_\delta f_0$, the bias $e = m - f_0$, and the sieve error $h := \widehat{f} - m \in \mathcal{F}_\delta - \mathcal{F}_\delta$,

$$\widehat{f} - f_0 = h + e.$$

Bias. By (14), $\|e\|_2^2 \lesssim \delta^{2\alpha}$.

Linearize about the comparison point (multiplicative r). Rather than linearize about $\mathbf{1}$ (which produces an *additive* remainder $\sim r\|h\|_2$ that cannot be absorbed at fixed r), we linearize about $m = \Pi_\delta f_0$. Note $1 - 2r \leq m \leq 1 + 2r$ and $\|\nabla u_m\|_\infty \lesssim 1$ ((T6), since $m \in \mathcal{F}(\alpha, 2r) \subset C^\alpha$). Write $u_{\widehat{f}} - u_m = T_m h + \widetilde{\rho}$, where T_m is the linearized forward map at m and $\widetilde{\rho}$ is its quadratic remainder. The same energy argument as Lemma 2.7, with ∇u_m in place of ∇u_0, gives $\|\widetilde{\rho}\|_2 \lesssim \|h\|_\infty \|h\|_2$. The linearization about $\mathbf{1}$, $T = T_{\mathbf{1}}$, differs from T_m by an operator-difference reflecting the r-closeness of m to $\mathbf{1}$; the essential point is that this enters *multiplicatively*: for wavelet-packet h on the sieve at scale δ the h-dependent part of $T_m - T$ satisfies

$$T_m h - Th = -\Delta^{-1} \operatorname{div}\big(h \nabla(u_m - u_0)\big), \qquad \|T_m h - Th\|_2 \;\lesssim\; r\,\delta\,\|h\|_2, \qquad m - \mathbf{1} = O(r) \text{ in } C^\alpha, \quad (18)$$

To see the identity in (18), subtract the linearized equations: $\operatorname{div}(\nabla T_m h) = -\operatorname{div}(h \nabla u_m)$ and $\operatorname{div}(\nabla T h) = -\operatorname{div}(h \nabla u_0)$ give $\operatorname{div} \nabla(T_m h - Th) = -\operatorname{div}\big(h \nabla(u_m - u_0)\big)$, i.e. $T_m h - Th = -\Delta^{-1} \operatorname{div}\big(h \nabla(u_m - u_0)\big)$ (this is a first-order expansion of the *difference* $(T_m - T)h$, the h-independent piece $(T_m - T)0 = u_m - u_0$ being exactly accounted for by $u_m - u_0 = \Delta^{-1} \operatorname{div}((m - \mathbf{1})\nabla u_0)$). For the bound, the only subtlety is the boundary layer of $m - \mathbf{1}$: by the interior Schauder estimate $\|u_m - u_0\|_{C^{2,\beta}(K)} \leq C_K \|m - \mathbf{1}\|_{C^\alpha}$ on compact $K \Subset D$ (the right side of $\operatorname{div}(m \nabla(u_m - u_0)) = \operatorname{div}((m - \mathbf{1})\nabla u_0)$ is the divergence of the vector field $(m - \mathbf{1})x/d$, whose $C^{\alpha-1}(K)$ norm is $\lesssim_K r$; (T6) applies) and the packet h is supported in $\{|x| \leq \frac{3}{4}\} =: K$, so $\|h \nabla(u_m - u_0)\|_2 \leq C_K r \|h\|_2$. Repeating the single-scale Fourier computation of Lemma 3.3 with the vector source $h \nabla(u_m - u_0)$ in place of xg — for the *upper* bound only, where the estimate reduces to $\|-\Delta^{-1} \operatorname{div} F\|_2 \lesssim \delta \|F\|_2$ for any vector field F whose spatial scale is δ (Lemma 3.3's upper bound (10) applied componentwise; the slow factor $\nabla(u_m - u_0)$ is $O(r)$ in L^∞ and $C^{\alpha-1}$ and does not change the scale) — yields

$$\|T_m h - Th\|_2 \;\lesssim\; \delta\, C_r\, \|h\|_2 \;\lesssim\; r\,\delta\,\|h\|_2$$

(this is the multiplicative r: the estimate scales linearly with $\|m - \mathbf{1}\|_{C^\alpha} \lesssim r$, and fails for $r = 1$). The remaining piece $(u_m - u_0)$ of $(T_m - T)$ is *independent of* h and is grouped, below, with the bias/discrepancy term (it acts as an additional $O(r)$ offset inside $u_m - u_{f_0}$; its contribution to the risk is handled with the bias in the accounting display below). Consequently the signal satisfies the recoverable lower bound

$$\|P_\delta T_m h\|_2 \;\geq\; \|P_\delta T h\|_2 - \|(T_m - T)h\|_2 \;\geq\; \big(c - Cr\big)\delta \|h\|_2 \;\geq\; \tfrac{c}{2}\,\delta \|h\|_2 \qquad \text{for } r \leq r_0 := \tfrac{c}{2C}, \quad (19)$$

by Lemma 5.2 applied to the annulus packet Th and (18). Since the sieve perturbations h are by construction supported in the bulk annulus (no origin-near modes are estimated; Lemma 5.2's lower

bound applies to the *whole* h), this is the recovery lower bound used in (22); on the origin content absorbed into Π_δ see Remark 5.4.

Absorption via per-scale Nikolskii on the compact sieve. On the compact sieve $\mathcal{F}_\delta \subset \{\|\cdot\|_{C^\alpha} \le 2r\}$, a difference $h = \sum_j h_j$ of two sieve elements has $\|h_j\|_{C^\alpha} \le 4r$ at each scale $2^{-j} \ge \delta$. The per-scale Nikolskii inequality and the C^α class cap give $\|h_j\|_\infty \lesssim \min\big(2^{jd/2}\|h_j\|_2,\, 2^{-j\alpha}r\big)$. Summing over the $\lesssim \log(1/\delta)$ scales $2^{-j} \ge \delta$, and using Lemma 2.7's $\|\widetilde{\rho}\|_2 \lesssim \|h\|_\infty \|h\|_2$ together with the pointwise triangle inequality $\|h\|_\infty \le \sum_j \|h_j\|_\infty$ (the pieces are summable in L^∞ since $\|h_j\|_\infty \lesssim 2^{-j\alpha}r$ and $\alpha > 0$),

$$\frac{\|\widetilde{\rho}\|_2}{\delta\|h\|_2} \;\le\; \frac{\|h\|_\infty}{\delta} \;\le\; \delta^{-1}\sum_{2^{-j}\ge\delta}\|h_j\|_\infty \;\lesssim\; \delta^{-1}\sum_{2^{-j}\ge\delta}\min\big(2^{jd/2}\|h_j\|_2,\, 2^{-j\alpha}r\big).$$

Using the bootstrap $\|h\|_2 \le 2r$ and the uniform C^α bound, each summand is $\lesssim 2^{-j\alpha}r$, so $\delta^{-1}\sum_j 2^{-j\alpha}r \lesssim \delta^{-1}r\delta^\alpha = r\delta^{\alpha-1} \to 0$ *because* $\alpha > 1$. Hence, at the scale of interest and quantitatively (no undefined exponent),

$$\|\widetilde{\rho}\|_2 \;\le\; r\,\delta^{\alpha-1}\big(\delta\|h\|_2\big) \;=:\; \omega_\delta\,\delta\|h\|_2, \qquad \omega_\delta := r\delta^{\alpha-1} \to 0 \;\; (\alpha > 1). \tag{20}$$

This is the genuine multiplicative absorption, valid for all $\alpha > 1$, all $d \ge 2$.

Basic inequality and risk. Write, abusing notation, $\Upsilon(g) = \|P_\delta u_g - Y_\delta\|_2^2$, $\Delta Y := P_\delta(u_{f_0} - Y_\delta) = n^{-1/2}\xi_\delta$ (the projected negative noise), and set

$$X \;:=\; P_\delta\big(u_{\widehat{f}} - u_m\big), \qquad Z \;:=\; P_\delta\big(u_{f_0} - u_m\big).$$

From $\Upsilon(\widehat{f}) \le \Upsilon(m)$, expanding the squares as $\|X - (\Delta Y + Z)\|_2^2 \le \|\Delta Y + Z\|_2^2$ and cancelling the common $\|\Delta Y + Z\|_2^2$,

$$\|X\|_2^2 \;\le\; 2\,\big|\langle X, \Delta Y + Z\rangle\big| \;\le\; 2\|X\|_2\big(\|\Delta Y\|_2 + \|Z\|_2\big),$$

so (dividing by $\|X\|_2$; if $X = 0$ the conclusion is trivial)

$$\|X\|_2 \;\le\; 2\,\|\Delta Y\|_2 + 2\,\|Z\|_2 \;=\; 2n^{-1/2}\|\xi_\delta\|_2 + 2\|P_\delta(u_{f_0} - u_m)\|_2. \tag{21}$$

On the left, $u_{\widehat{f}} - u_m = T_m h + \widetilde{\rho}$ with $\|\widetilde{\rho}\|_2 \le \omega_\delta\,\delta\|h\|_2$ by (20), hence by (19)

$$\|X\|_2 \;\ge\; \|P_\delta T_m h\|_2 - \|\widetilde{\rho}\|_2 \;\ge\; \tfrac{c}{2}\delta\|h\|_2 - \omega_\delta\delta\|h\|_2 \;\ge\; \tfrac{c}{4}\,\delta\|h\|_2 \qquad (\delta \le \delta_0).$$

For the right side of (21), the *discrepancy* $u_{f_0} - u_m$ is the solution error generated by the approximation residual $f_0 - m = f_0 - \Pi_\delta f_0 = -e$; by the fine-scale bias bound (next paragraph), $\|u_{f_0} - u_m\|_2 \lesssim \delta^{1+\alpha}$, i.e. $\|Z\|_2 = \|P_\delta(u_{f_0} - u_m)\|_2 \lesssim \delta^{1+\alpha}$. Combining the last two displays with (21),

$$\tfrac{c}{4}\delta\|h\|_2 \;\lesssim\; n^{-1/2}\|\xi_\delta\|_2 + \delta^{1+\alpha} + \omega_\delta\,\delta\|h\|_2, \tag{22}$$

and the $\omega_\delta\delta\|h\|_2$ term is absorbed on the left for $\delta \le \delta_0$ (since $\omega_\delta \to 0$).

Fine-scale bias. Write $w := u_{f_0} - u_m$; both u_{f_0} and u_m vanish on ∂D (the Dirichlet condition of the very-weak formulation (4)), so $w \in H_0^1$ solves $\operatorname{div}(f_0\nabla w) = \operatorname{div} G$, $G := e\nabla u_m$, with $\|G\|_2 \le \|e\|_2\|\nabla u_m\|_\infty \lesssim \delta^\alpha$ by (14) and $\|\nabla u_m\|_\infty \lesssim 1$ (T6). Split $G = G_{\text{int}} + G_{\text{col}}$, $G_{\text{col}} = G\,\mathbf{1}_S$ the part in a collar $S = \{|x| > 1 - s\}$ of width s around ∂D. Two points: (i) *collar remainder*: by Hölder and (14), $\|e\|_{L^2(S)} \le \|e\|_{L^p(S)}|S|^{\frac{1}{2}-\frac{1}{p}} \lesssim s^{\frac{1}{2}-\frac{1}{p}}\delta^\alpha$, with $p > 2$ taken large (the C^α-cap tail is Nikolskii-small, $\|e\|_{L^p(S)} \lesssim \delta^\alpha$, the projected region $O(r)$); for $s \lesssim \delta$ the collar is $\lesssim \delta^{\alpha+\frac{1}{4}} = o(\delta^\alpha)$, and no $r\delta^\alpha$ term appears — collar content is projected into Π_δ; (ii) *interior gain*: each scale-δ bump of e, at distance $\gtrsim s$ from ∂D, is mean-zero, so by the exact Fourier mechanism of Lemma 3.3 (the $|\xi|^{-1}$-gain on the

loaded shell; the harmonic correction (11) applies since supp G_{int} is interior) $\|G_{\mathrm{int}}\|_{H^{-1}} \lesssim \delta\|e\|_2 \lesssim \delta^{\alpha+1}$. Combining, testing against $\phi = -\Delta_{f_0}^{-1}\psi$ as in Lemma 2.6 and using Poincaré on $w \in H_0^1$,

$$\|w\|_2 = \|u_{f_0} - u_m\|_2 \;\lesssim\; \|G\|_{H^{-1}} \;\lesssim\; \delta\|e\|_2 + \delta^{\alpha+\frac{1}{4}} \;\lesssim\; \delta^{1+\alpha}. \tag{23}$$

Thus $\|Z\|_2 = \|P_\delta(u_{f_0} - u_m)\|_2 \lesssim \delta^{1+\alpha}$.

Noise enters linearly (no union bound necessary). In the comparison inequality the noise appears only through the *linear* functional $h \mapsto n^{-1/2}\langle \xi_\delta, P_\delta T h\rangle$ evaluated at the random point $h = \widehat{f} - \Pi_\delta f_0$; by Cauchy–Schwarz, $n^{-1/2}|\langle \xi_\delta, P_\delta T h\rangle| \le n^{-1/2}\|\xi_\delta\|_2 \, \|P_\delta T h\|_2$, and after division by $\|P_\delta T h\|_2$ this contributes $n^{-1/2}\|\xi_\delta\|_2$ on the event $\{\widehat{f} \in \mathcal{F}_\delta\}$, uniformly. Because the projected model is finite-dimensional and the functional is linear, $\mathbb{E}\|\xi_\delta\|_2^2 = \dim V_\delta \asymp \delta^{-d}$, and *no supremum / union bound over the sieve is needed here*: a logarithmic factor would arise only if one insisted on a *uniform-in-g* comparison, which the least-squares basic inequality avoids. Account for every term of (22): after absorbing $\omega_\delta\delta\|h\|_2$ and dividing by δ, $\|h\|_2 \lesssim \delta^{-1}\big(n^{-1/2}\|\xi_\delta\|_2 + \delta^{1+\alpha}\big)$, so squaring (and using $\mathbb{E}\|\xi_\delta\|_2^2 \asymp \delta^{-d}$) $\mathbb{E}\|h\|_2^2 \lesssim \delta^{-d-2}/n + \delta^{2\alpha}$: the discrepancy $\delta^{1+\alpha}$ contributes $\delta^{2\alpha}$, not $r^2\delta^{2\alpha-2}$ — the boundary collar/collar content responsible for the latter has been projected into the bias, whose u-image uses the interior degree-one gain (23). With the balance $\delta = \delta_* = n^{-1/(2\alpha+d+2)}$ chosen so $\delta_*^{-d-2}/n = \delta_*^{2\alpha}$, one therefore obtains the *log-free* estimate

$$\mathbb{E}\|\widehat{f} - f_0\|_2^2 \lesssim \|e\|_2^2 + \mathbb{E}\|h\|_2^2 \lesssim \delta_*^{2\alpha} + \frac{\delta_*^{-d-2}}{n} \asymp n^{-\frac{2\alpha}{2\alpha+d+2}}.$$

All contributing terms (bias, discrepancy, noise) are $\lesssim \delta_*^{2\alpha}$ at the balance; the upper bound is free of logarithmic loss.

6 Conclusion and consistency checks

- **Rate.** Lower bound (Proposition 4.1): $n^{-2\alpha/(2\alpha+d+2)}$ (no log). Upper bound (Proposition 5.1): $n^{-2\alpha/(2\alpha+d+2)}$ (no log: the noise enters the least-squares basic inequality only through a *linear* functional, so no union bound is needed). The two agree, so the minimax rate is $n^{-2\alpha/(2\alpha+d+2)}$ up to logarithmic factors.
- **Interpretation.** The exponent denominator $2\alpha + d + 2$ exceeds the direct-problem $2\alpha + d$ by exactly 2, reflecting *one* degree of smoothing of the linearized inverse T (order -1), a consequence of the transport structure $\Delta T = I + \frac{x\cdot\nabla}{d}$ and the L^2-invertibility of $I + \frac{x\cdot\nabla}{d}$ on $L^2(D)$ (Lemmas 2.1, 2.3 and the exact Fourier identity Lemma 3.1).
- **Role of $\alpha > 1$.** Used decisively in both bounds to make the quadratic remainder $R(h)$ (Lemma 2.7) negligible against the linear signal Th at the minimax scale (ratio $\asymp \delta_*^{\alpha-1} \to 0$), cf. Remark 2.8. For $\alpha \le 1$ the linearization is not faithful at scale δ_*; the rate would then require a genuinely nonlinear analysis (we make no claim in that regime).
- **Role of small r.** $r \le r_0(\alpha, d)$ is used for (i) uniform ellipticity $1 - r \le f \le 1 + r$, (ii) absorbing the quadratic remainder into the linear signal, and (iii) the uniform apriori bound $\|u_f\|_\infty \lesssim 1$. All structural constants and r_0 depend only on (α, d, ψ); C is uniform in $r \in (0, r_0]$ and n; the lower-bound c carries the r^2 factor of the shrinking class.
- **Model handling.** u_f is defined by the very-weak formulation (4); no H^1 membership is asserted (Section 2.5). The white noise is only ever projected onto the finite-dimensional sieve V_δ, where it is a well-defined Gaussian vector, so the estimator/risk are rigorous on the white-noise model.